

Prediction of sugarcane prices in Pakistan using Time Series Methods

1 Problem Statement

The aim of this assignment is to analyze and forecast sugar prices in Pakistan using a time series dataset spanning approximately 13 years. The study examines how sugar prices have varied over time and aims to predict future trends based on historical data.

Significance of Forecasting Sugar Prices in Pakistan

Forecasting sugar prices in Pakistan is crucial due to the country's heavy reliance on sugarcane as a key agricultural commodity and the integral role sugar plays in both consumer markets and industrial production. Accurate price forecasting enables farmers, mill owners, policymakers, and businesses to make informed decisions regarding crop cultivation, procurement strategies, inventory management, and pricing regulations.

Moreover, with inflation and economic instability affecting food security, reliable sugar price predictions can aid in designing targeted consumer protection policies to mitigate the impact of price surges on households.

2 Data Collection

The sugar price data used in this study was collected from the Karandaaz data portal, specifically from the dataset titled "Sugar Price" available at <https://portal.karandaaz.com.pk/dataset/sugar-refined/1361>. The dataset provides historical and current sugar price information relevant to Pakistan being updated on a weekly basis. The dataset in focus here provides prices for the period of March 2013 to March 2025.

The dataset includes sugar prices from various cities across Pakistan, such as Karachi, Lahore, Islamabad, and other major cities. This attribute helps analyze how sugar prices vary geographically within the country.

The dataset also provides the average sugar price across Pakistan, offering a consolidated view of price trends at the national level. This attribute is crucial for forecasting and policy-making at a macroeconomic level.

3 Importing the required libraries

```
[1]: import numpy as np
import pandas as pd
import os
import seaborn as sns
```

```
[9]: # from google.colab import drive
# drive.mount('/content/drive')
```

```
[10]: # # Get the current working directory
# current_directory = os.getcwd()
# print(f"Current directory: {current_directory}")
# os.chdir('/content/drive/MyDrive/Colab Notebooks/TSAF Assignment')
# current_directory = os.getcwd()
# print(f"Current directory: {current_directory}")
```

4 Loading the dataset

```
[3]: suger_prices_2013_2025_pak = pd.read_csv("Sugar Price.csv")
```

```
[13]: # #suger_prices_2013_2025_pak = pd.read_csv("current_directory/Sugar Price.csv")
# suger_prices_2013_2025_pak = pd.read_csv("Sugar Price.csv")
# )
```

```
[5]: suger_prices_2013_2025_pak.head()
```

```
[5]:
```

	Period End	Islamabad	Rawalpindi	Gujranwala	Sialkot	Lahore	\
0	Weekly	PKR	PKR	PKR	PKR	PKR	
1	13-Mar-2025	176.240000	177.650000	175	175	172.940000	
2	07-Mar-2025	169.410000	169.660000	165	165	163.320000	
3	27-Feb-2025	163.740000	161.650000	160	160	160	
4	20-Feb-2025	161.240000	159.660000	155	158	156.410000	

	Faisalabad	Sargodha	Multan	Bahawalpur	Karachi	Hyderabad	Sukkur	\
0	PKR	PKR	PKR	PKR	PKR	PKR	PKR	
1	170	170	168.990000	170	174.370000	170	168	
2	160	160	158.990000	160	162.540000	160	158	
3	155	155	156.420000	155	157.100000	156.650000	155	
4	155	155	154.200000	153.320000	153.280000	150	150	

	Larkana	Peshawar	Bannu	Quetta	Khuzdar	Pakistan
0	PKR	PKR	PKR	PKR	PKR	PKR
1	168.320000	171.100000	170	174.490000	170.670000	171.900000
2	158.320000	166.650000	160	165.990000	162.650000	162.640000
3	153.320000	160.550000	155	158.990000	157.650000	157.680000
4	150	160.550000	155	157.490000	155.990000	155.270000

5 Exploring and cleaning the dataset

```
[16]: # Inspecting the column data types:
suger_prices_2013_2025_pak.dtypes
```

```
[16]: Period End      object
Islamabad      object
Rawalpindi     object
Gujranwala     object
Sialkot        object
Lahore         object
Faisalabad     object
Sargodha       object
Multan         object
Bahawalpur     object
Karachi        object
Hyderabad     object
Sukkur         object
Larkana        object
Peshawar       object
Bannu          object
Quetta        object
Khuzdar        object
Pakistan       object
dtype: object
```

The date and the prices column are of the 'object' data type which will not be suitable for analysis. So converting these columns to the relevant datatypes

```
[11]: suger_prices_2013_2025_pak_cleaned = suger_prices_2013_2025_pak.iloc[1:].copy()
suger_prices_2013_2025_pak_cleaned.head()
```

```
[11]:
```

	Period End	Islamabad	Rawalpindi	Gujranwala	Sialkot	Lahore	\
1	13-Mar-2025	176.240000	177.650000	175	175	172.940000	
2	07-Mar-2025	169.410000	169.660000	165	165	163.320000	
3	27-Feb-2025	163.740000	161.650000	160	160	160	
4	20-Feb-2025	161.240000	159.660000	155	158	156.410000	
5	13-Feb-2025	158.740000	158.650000	160	160	156.410000	

	Faisalabad	Sargodha	Multan	Bahawalpur	Karachi	Hyderabad	Sukkur	\
1	170	170	168.990000	170	174.370000	170	168	
2	160	160	158.990000	160	162.540000	160	158	
3	155	155	156.420000	155	157.100000	156.650000	155	
4	155	155	154.200000	153.320000	153.280000	150	150	
5	150	152	154.200000	151.650000	154.040000	150	150	

	Larkana	Peshawar	Bannu	Quetta	Khuzdar	Pakistan
1	168.320000	171.100000	170	174.490000	170.670000	171.900000

2	158.320000	166.650000	160	165.990000	162.650000	162.640000
3	153.320000	160.550000	155	158.990000	157.650000	157.680000
4	150	160.550000	155	157.490000	155.990000	155.270000
5	150	156.650000	150	154.990000	155.990000	154.270000

```
[31]: # Converting the 'period_end' column to the 'date' data type:
suger_prices_2013_2025_pak_cleaned['Period End'] = pd.
    ↳to_datetime(suger_prices_2013_2025_pak_cleaned['Period End'],
    ↳format='%d-%b-%Y')
suger_prices_2013_2025_pak_cleaned['Period End'].head()
```

```
[31]: 1    2025-03-13
      2    2025-03-07
      3    2025-02-27
      4    2025-02-20
      5    2025-02-13
      Name: Period End, dtype: datetime64[ns]
```

```
[35]: # Converting all the columns with prices to 'numeric' data type:
price_columns = suger_prices_2013_2025_pak_cleaned.columns[1:]
for col in price_columns:
    suger_prices_2013_2025_pak_cleaned[col] = pd.
    ↳to_numeric(suger_prices_2013_2025_pak_cleaned[col], errors='coerce')

suger_prices_2013_2025_pak_cleaned.dtypes
```

```
[35]: Period End      datetime64[ns]
      Islamabad      float64
      Rawalpindi     float64
      Gujranwala     float64
      Sialkot        float64
      Lahore         float64
      Faisalabad     float64
      Sargodha       float64
      Multan         float64
      Bahawalpur     float64
      Karachi       float64
      Hyderabad     float64
      Sukkur        float64
      Larkana       float64
      Peshawar      float64
      Bannu         float64
      Quetta        float64
      Khuzdar       float64
      Pakistan      float64
      dtype: object
```

5.1 Now checking for any null values present in the data

```
[17]: suger_prices_2013_2025_pak_cleaned.isnull().any()
```

```
[17]: Period End      False
      Islamabad     False
      Rawalpindi    False
      Gujranwala    False
      Sialkot       False
      Lahore        False
      Faisalabad    False
      Sargodha      False
      Multan        False
      Bahawalpur    False
      Karachi      False
      Hyderabad    False
      Sukkur        False
      Larkana       False
      Peshawar      False
      Bannu         False
      Quetta        False
      Khuzdar       False
      Pakistan     False
      dtype: bool
```

no null values found in any columns of the loaded dataset

5.2 Checking for any duplicates in the dataset

```
[25]: suger_prices_2013_2025_pak_cleaned.duplicated()
```

```
[25]: 1      False
      2      False
      3      False
      4      False
      5      False
      ...
      570    False
      571    False
      572    False
      573    False
      574    False
      Length: 574, dtype: bool
```

no duplicates values found

5.3 Checking the overall dimensions of the data and other relevant summary information

```
[28]: suger_prices_2013_2025_pak_cleaned.shape
```

```
[28]: (574, 19)
```

```
[29]: suger_prices_2013_2025_pak_cleaned.describe()
```

```
[29]:
```

		Period End	Islamabad	Rawalpindi	Gujranwala	\
count		574	574.000000	574.000000	574.000000	
mean	2019-05-29 13:30:18.815331072		87.985279	85.535348	83.433101	
min	2013-01-03 00:00:00		55.500000	52.250000	50.000000	
25%	2016-08-05 18:00:00		60.750000	57.872500	55.500000	
50%	2019-08-12 00:00:00		77.750000	73.980000	72.000000	
75%	2022-05-31 06:00:00		100.000000	98.895000	96.997500	
max	2025-03-13 00:00:00		186.200000	192.980000	180.000000	
std		NaN	31.725429	32.685041	32.000626	

	Sialkot	Lahore	Faisalabad	Sargodha	Multan	Bahawalpur	\
count	574.000000	574.000000	574.000000	574.000000	574.000000	574.000000	
mean	83.364199	84.442282	82.208397	81.690209	81.374582	82.226725	
min	50.000000	51.790000	49.000000	48.000000	48.000000	50.000000	
25%	56.000000	57.640000	55.312500	55.000000	54.500000	55.000000	
50%	72.250000	74.140000	72.000000	70.500000	70.885000	72.250000	
75%	95.000000	95.700000	95.000000	95.000000	94.330000	95.000000	
max	180.000000	180.470000	170.780000	175.670000	175.990000	175.000000	
std	31.970286	31.724704	31.139217	31.231237	31.173733	30.849539	

	Karachi	Hyderabad	Sukkur	Larkana	Peshawar	Bannu	\
count	574.000000	574.000000	574.000000	574.000000	574.000000	574.000000	
mean	84.120714	81.523815	80.340767	81.679826	84.738240	82.316638	
min	50.620000	49.000000	48.000000	47.500000	50.330000	49.500000	
25%	57.190000	55.000000	54.000000	55.000000	57.330000	55.500000	
50%	72.555000	70.000000	70.000000	70.000000	73.995000	72.250000	
75%	99.300000	96.077500	95.000000	95.000000	99.382500	95.000000	
max	183.140000	180.000000	180.000000	180.000000	188.790000	180.000000	
std	31.976215	31.127753	30.243164	31.153722	32.582988	30.850327	

	Quetta	Khuzdar	Pakistan
count	574.000000	574.000000	574.000000
mean	86.269077	82.982735	83.273589
min	51.000000	50.000000	50.560000
25%	57.000000	55.000000	55.807500
50%	75.000000	72.000000	72.440000
75%	100.000000	95.247500	96.665000
max	197.990000	188.660000	181.980000
std	34.968179	32.571609	31.645498

The data has 574 entries for per kg prices of sugar from weeks starting from 03-Jan-2013 to 13-Mar-2025 across 18 cities in Pakistan with the 19th column indicating the overall prices in the country as a whole.

The mean sugar price per kg is not highly different across the cities during the time period, with Islamabad having the highest average price at around 87.98 PKR per kg and Sukkur having the lowest average sugar price at 80.34 PKR per kg. The difference between the highest and lowest mean values is around 7.64, which is a moderate variation.

The standard deviations across cities are roughly 30-32, indicating a similar spread of data points around the mean.

The 25th percentile (Q1) and 75th percentile (Q3) for sugar prices values vary across cities, but not drastically.

The 75th percentile for Karachi (PKR 99.3) is slightly higher than for some other cities, but overall, the IQRs do not indicate extreme variations.

The minimum prices are relatively close across cities (around PKR 48-55), but the maximum prices show larger variation (from PKR 170.78 to 192.98), indicating price spikes in some cities.

Cities close to production areas (e.g., Lahore, Faisalabad, Multan, Hyderabad, Sukkur) might have lower prices due to reduced transportation costs and higher local supply.

Cities further away (e.g., Quetta) might experience higher prices due to transportation and distribution costs.

Karachi, being a major commercial hub with high demand, often experiences higher prices than production hubs like Sukkur or Faisalabad.

```
[37]: suger_prices_2013_2025_pak_cleaned['Pakistan'].describe()
```

```
[37]: count    574.000000
      mean     83.273589
      std     31.645498
      min     50.560000
      25%     55.807500
      50%     72.440000
      75%     96.665000
      max    181.980000
      Name: Pakistan, dtype: float64
```

```
[37]: # Checking the trend of yearly mean and median sugar prices in Pakistan
      suger_prices_2013_2025_pak_cleaned['Year'] =
      ↳ suger_prices_2013_2025_pak_cleaned['Period End'].dt.year
```

```
suger_prices_2013_2025_pak_cleaned['Month'] =
    →suger_prices_2013_2025_pak_cleaned['Period End'].dt.month
suger_prices_2013_2025_pak_cleaned['Month Name'] =
    →suger_prices_2013_2025_pak_cleaned['Period End'].dt.month_name()
suger_prices_2013_2025_pak_cleaned.head()
# yearly_sugar_prices_pk = suger_prices_2013_2025_pak_cleaned.
    →groupby('Year')['Pakistan']
# yearly_avg_and_median_sugar_prices_pk = yearly_sugar_prices_pk.agg(['mean',
    →'median'])
# yearly_avg_and_median_sugar_prices_pk
```

```
[37]: Period End  Islamabad  Rawalpindi  Gujranwala  Sialkot  Lahore  Faisalabad  \
1  2025-03-13      176.24      177.65      175.0      175.0      172.94      170.0
2  2025-03-07      169.41      169.66      165.0      165.0      163.32      160.0
3  2025-02-27      163.74      161.65      160.0      160.0      160.00      155.0
4  2025-02-20      161.24      159.66      155.0      158.0      156.41      155.0
5  2025-02-13      158.74      158.65      160.0      160.0      156.41      150.0
```

```
      Sargodha  Multan  Bahawalpur  ...  Sukkur  Larkana  Peshawar  Bannu  \
1      170.0  168.99      170.00  ...  168.0      168.32      171.10  170.0
2      160.0  158.99      160.00  ...  158.0      158.32      166.65  160.0
3      155.0  156.42      155.00  ...  155.0      153.32      160.55  155.0
4      155.0  154.20      153.32  ...  150.0      150.00      160.55  155.0
5      152.0  154.20      151.65  ...  150.0      150.00      156.65  150.0
```

```
      Quetta  Khuzdar  Pakistan  Year  Month  Month Name
1  174.49  170.67      171.90  2025      3      March
2  165.99  162.65      162.64  2025      3      March
3  158.99  157.65      157.68  2025      2  February
4  157.49  155.99      155.27  2025      2  February
5  154.99  155.99      154.27  2025      2  February
```

[5 rows x 22 columns]

Both the mean and the median yearly sugar prices are relatively close. This shows that there aren't excessive fluctuations or extreme price variations within each year. If there were large spikes or dips in sugar prices, the mean would be significantly different from the median.

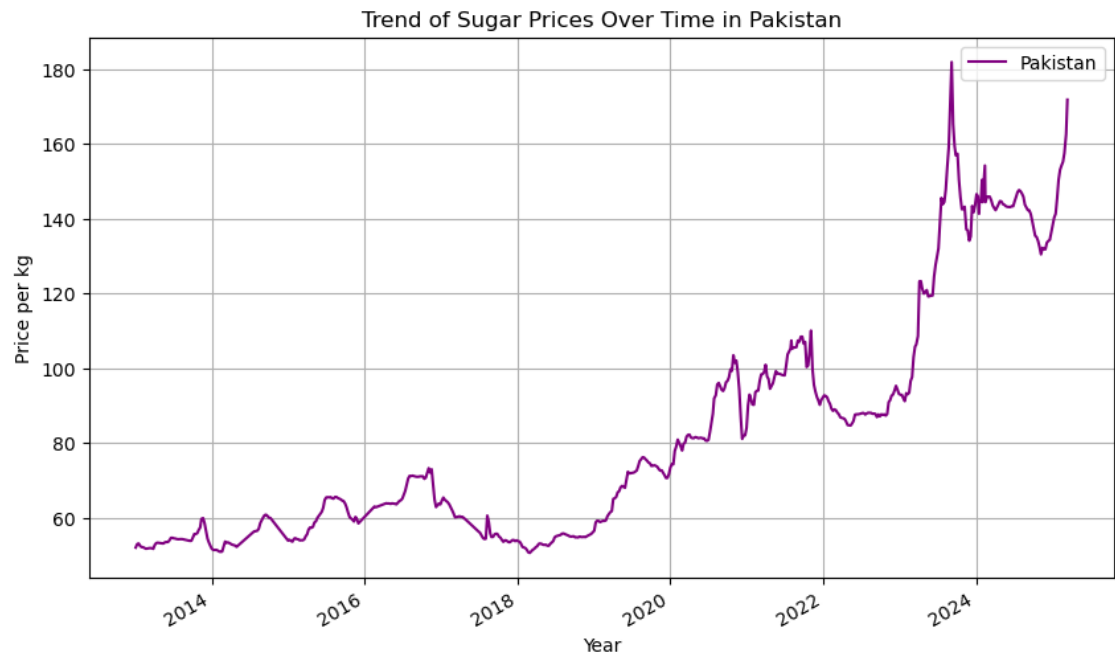
5.4 Visualizing the overall trend of sugar prices in Pakistan

```
[41]: # Plot the 'Pakistan' column over time
suger_prices_2013_2025_pak_cleaned.plot(
    x='Period End',
    y='Pakistan',
    color='purple',
    legend=True,
```



```
grid=True,
title='Trend of Sugar Prices Over Time in Pakistan',
xlabel='Year',
ylabel='Price per kg',
figsize=(10, 6) # Adjust the figure size if needed
)
```

[41]: <Axes: title={'center': 'Trend of Sugar Prices Over Time in Pakistan'},
xlabel='Year', ylabel='Price per kg'>



From 2013 to 2025, it can be seen that sugar prices have been consistently rising over time. The sharp overall increase after 2020 could indicate economic changes like inflation, supply chain disruptions, or policy shifts.

```
[45]: # Group by Year and calculate mean, median, variance, and standard deviation
price_stats = suger_prices_2013_2025_pak_cleaned.groupby('Year')['Pakistan'].
    .agg(['mean', 'median', 'var', 'std'])
price_stats = price_stats.reset_index()
# Display the results
print(price_stats)
```

	Year	mean	median	var	std
0	2013	53.961556	53.480	3.879827	1.969728
1	2014	54.991250	53.335	12.708915	3.564956
2	2015	59.759184	59.810	18.998333	4.358708
3	2016	67.343158	66.145	13.920498	3.731018

4	2017	57.616744	55.700	15.408437	3.925358
5	2018	53.798824	54.700	2.401179	1.549574
6	2019	68.531837	71.380	38.202232	6.180795
7	2020	86.327925	81.930	67.968598	8.244307
8	2021	98.818000	98.395	33.438000	5.782560
9	2022	88.731765	87.850	6.526431	2.554688
10	2023	129.027647	132.070	546.879982	23.385465
11	2024	142.114808	143.505	27.942606	5.286077
12	2025	153.251000	153.690	92.950766	9.641098

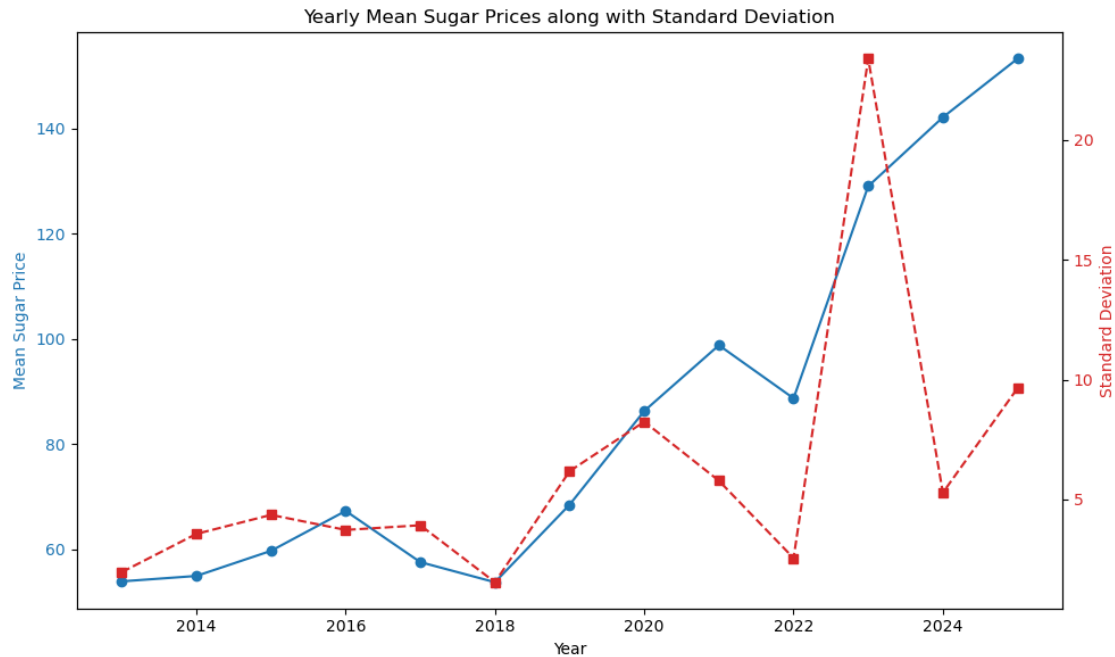
```
[39]: import matplotlib.pyplot as plt
import seaborn as sns
```

```
[47]: fig, ax1 = plt.subplots(figsize=(10, 6))

# First y-axis (Mean Prices)
ax1.set_xlabel("Year")
ax1.set_ylabel("Mean Sugar Price", color="tab:blue")
ax1.plot(price_stats['Year'] , price_stats['mean'], marker="o", color="tab:
↪blue", label="Mean Price per kg")
ax1.tick_params(axis="y", labelcolor="tab:blue")

# Second y-axis (Standard Deviation)
ax2 = ax1.twinx()
ax2.set_ylabel("Standard Deviation", color="tab:red")
ax2.plot(price_stats['Year'],price_stats['std'], marker="s", linestyle="dashed",
↪color="tab:red", label="Std Deviation")
ax2.tick_params(axis="y", labelcolor="tab:red")

# Titles & Legends
plt.title("Yearly Mean Sugar Prices along with Standard Deviation")
fig.tight_layout()
plt.show()
```



Before 2019, both the mean price and standard deviation were relatively low and stable. This suggests that sugar prices were more predictable in this period, with no major disruptions affecting supply or demand.

From 2020 to 2023, sugar prices significantly increased. Standard deviation peaked in 2023, indicating high price fluctuations (instability in sugar prices). Economic instability could be one of the reasons affecting such commodity prices.

After 2023, standard deviation drops significantly, even though prices continue rising. This suggests that sugar prices are increasing more steadily, without extreme fluctuations. It could indicate market stabilization due to better supply chain management, policy adjustments, or steady demand.

5.5 Visualizing the monthly trend of sugar prices in Pakistan by Year

```
[204]: # Create a custom order for months
month_order = ['January', 'February', 'March', 'April', 'May', 'June',
               'July', 'August', 'September', 'October', 'November', 'December']

for year in range(2013, 2025):
    # Filter the data for the specific year
    yearly_sugar_prices_pk =
    ↳ suger_prices_2013_2025_pak_cleaned[suger_prices_2013_2025_pak_cleaned['Year']
    ↳ == year]

    # Convert 'Month' to a categorical type with the specified order
```

```

yearly_sugar_prices_pk['Month'] = pd.
→Categorical(yearly_sugar_prices_pk['Month'], categories=month_order,
→ordered=True)

# Sort the DataFrame by the ordered 'Month' column
yearly_sugar_prices_pk = yearly_sugar_prices_pk.sort_values('Month')

# Set up the figure and axes
plt.figure(figsize=(10, 5))

# Plot the line graph with a solid color
plt.plot(yearly_sugar_prices_pk['Month Name'],
→yearly_sugar_prices_pk['Pakistan'], color='blue', linewidth=2)

# # Generate rainbow colors for the line segments based on the number of
→months
# colors = plt.cm.rainbow(np.linspace(0, 1, len(yearly_sugar_prices_pk)))

# # Plot the line graph with rainbow-colored segments
# for i in range(1, len(yearly_sugar_prices_pk)):
#     plt.plot(yearly_sugar_prices_pk['Month Name'].iloc[i-1:i+1],
→yearly_sugar_prices_pk['Pakistan'].iloc[i-1:i+1],
#             color=colors[i-1], linewidth=2)

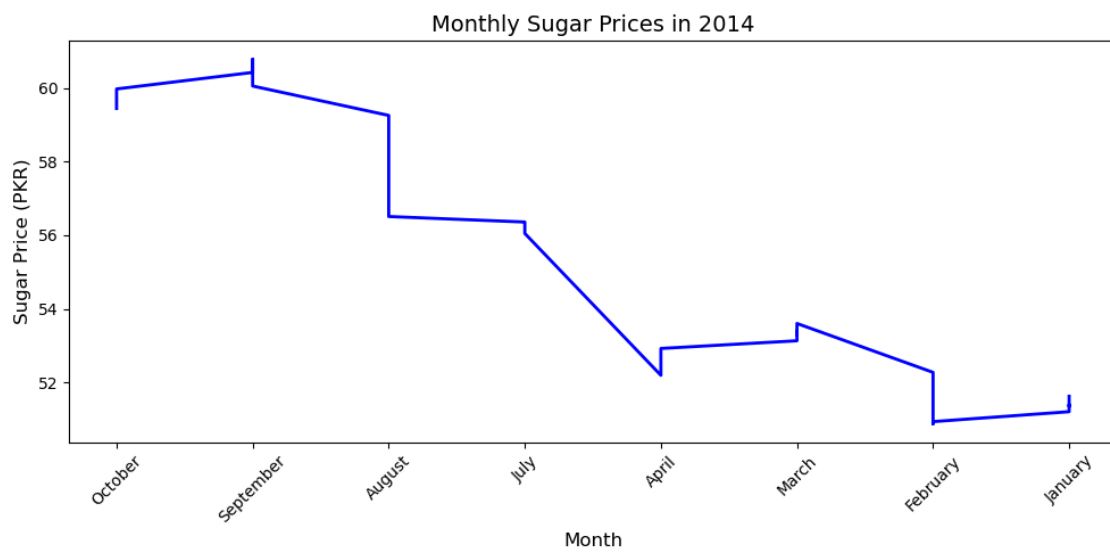
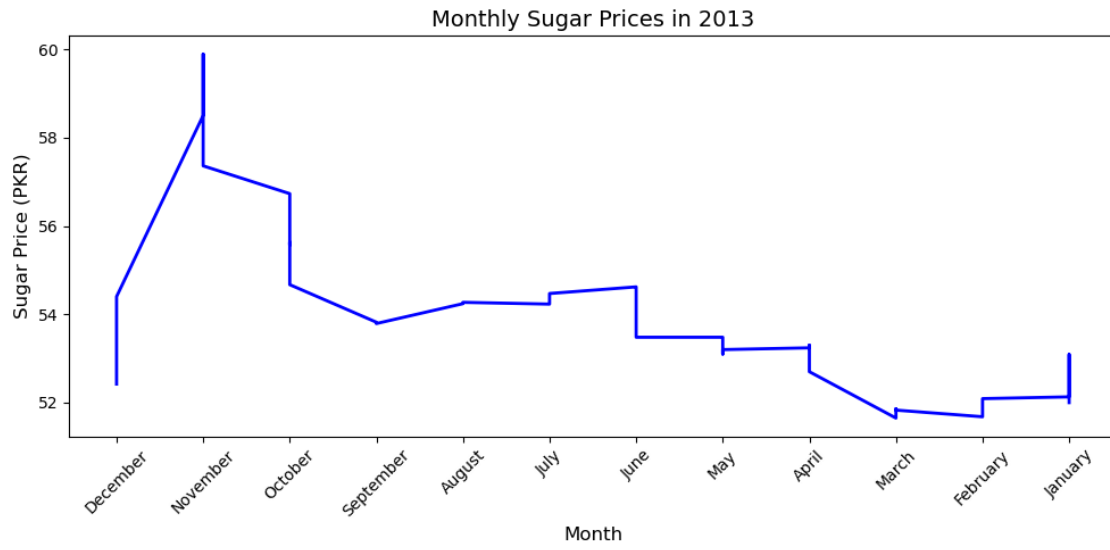
# Add values on each month point
# for i, (month, price) in enumerate(zip(yearly_sugar_prices_pk['Month'],
→yearly_sugar_prices_pk['Pakistan'])):
#     plt.text(i, price, f'{price:.2f}', ha='center', va='bottom',
→fontsize=10)

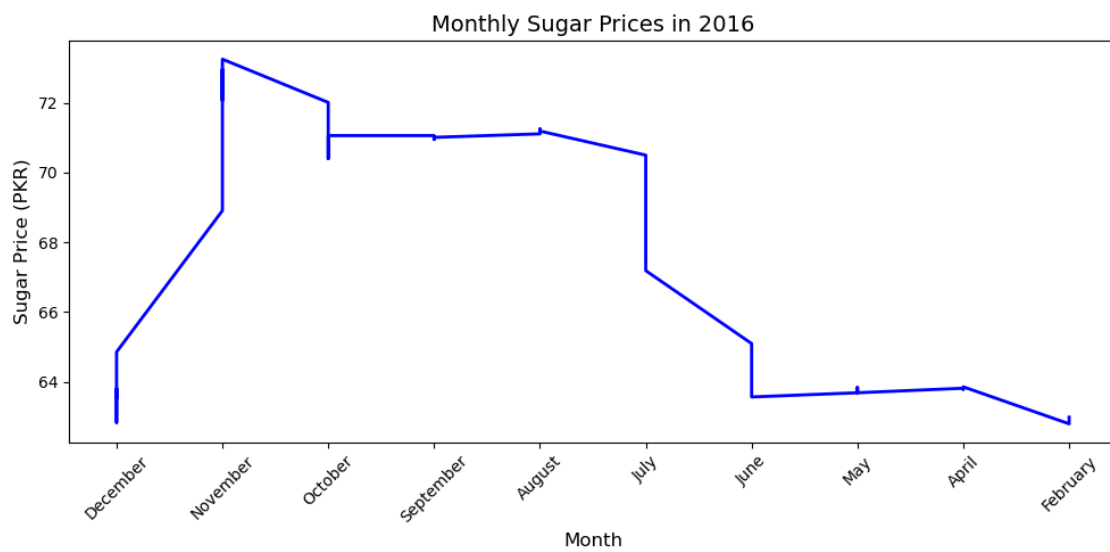
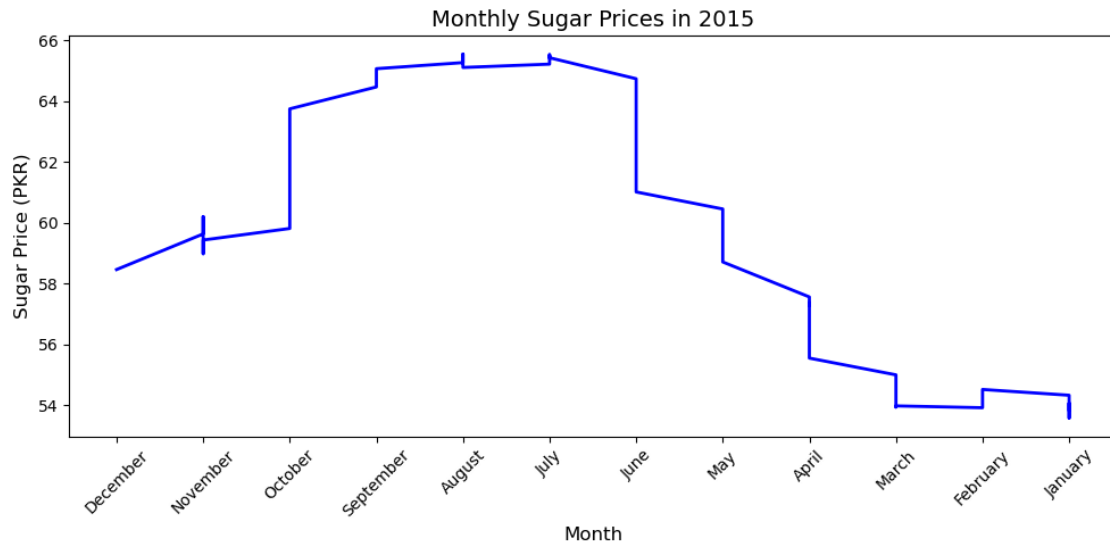
# Set the title and labels
plt.title(f'Monthly Sugar Prices in {year}', fontsize=14)
plt.xlabel('Month', fontsize=12)
plt.ylabel('Sugar Price (PKR)', fontsize=12)

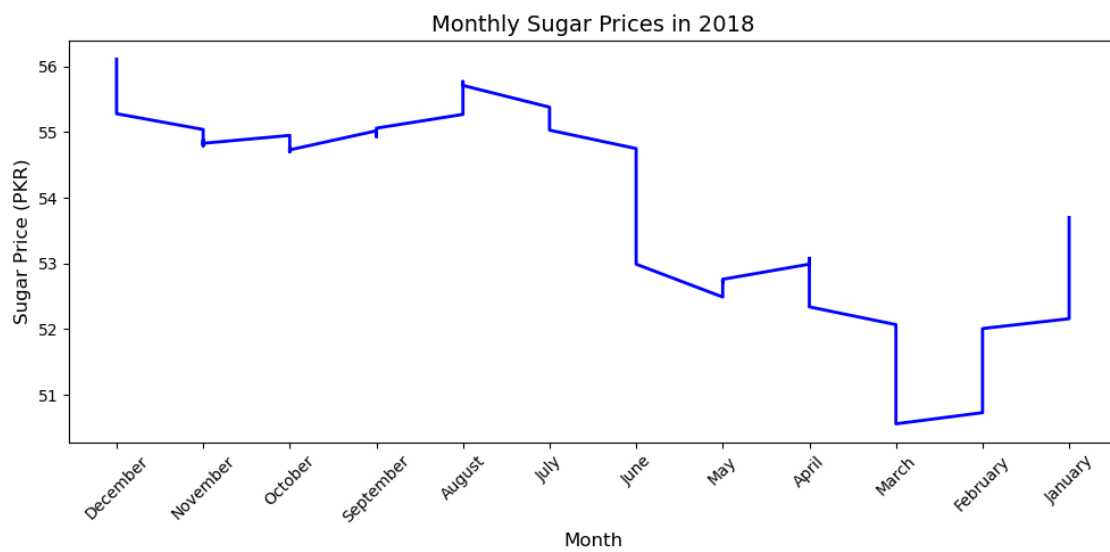
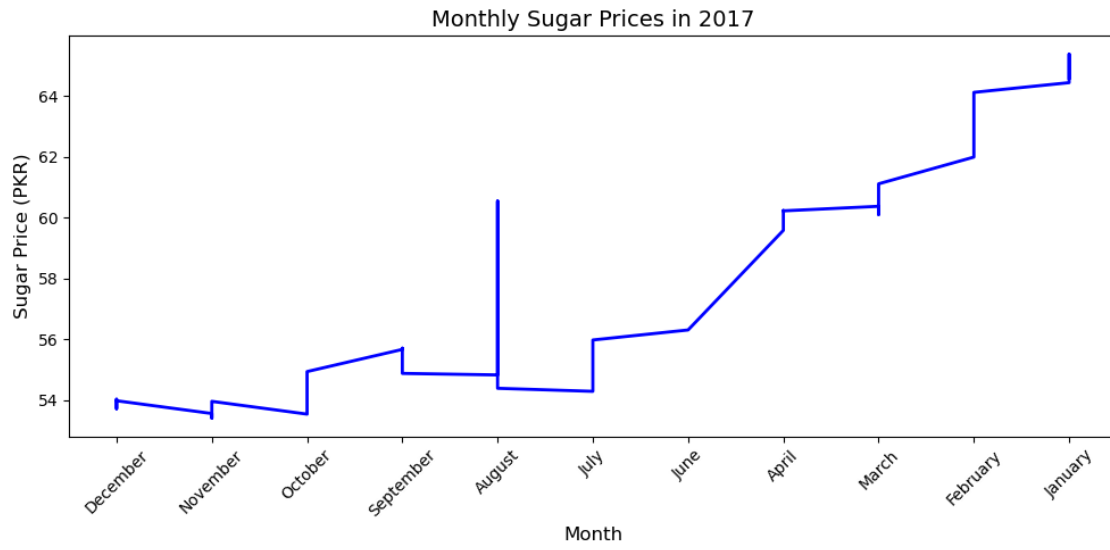
# Rotate month names for better readability
plt.xticks(rotation=45)

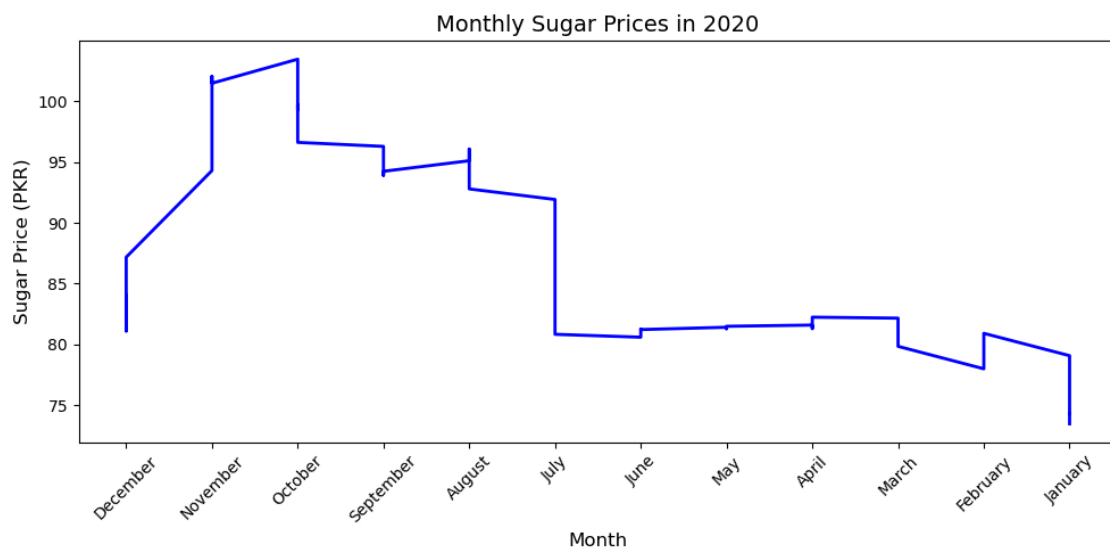
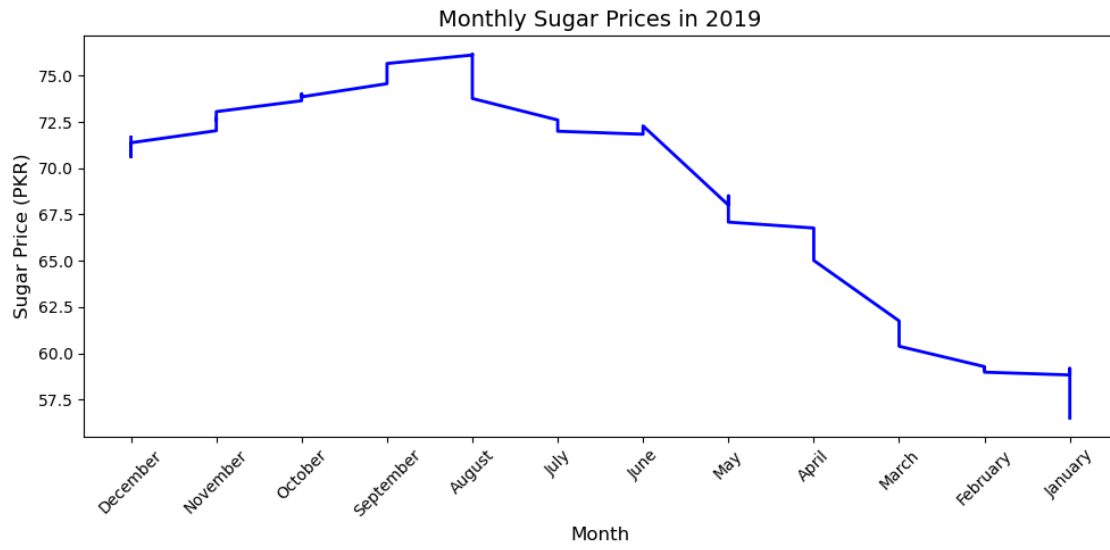
# Display the plot
plt.tight_layout()
plt.show()

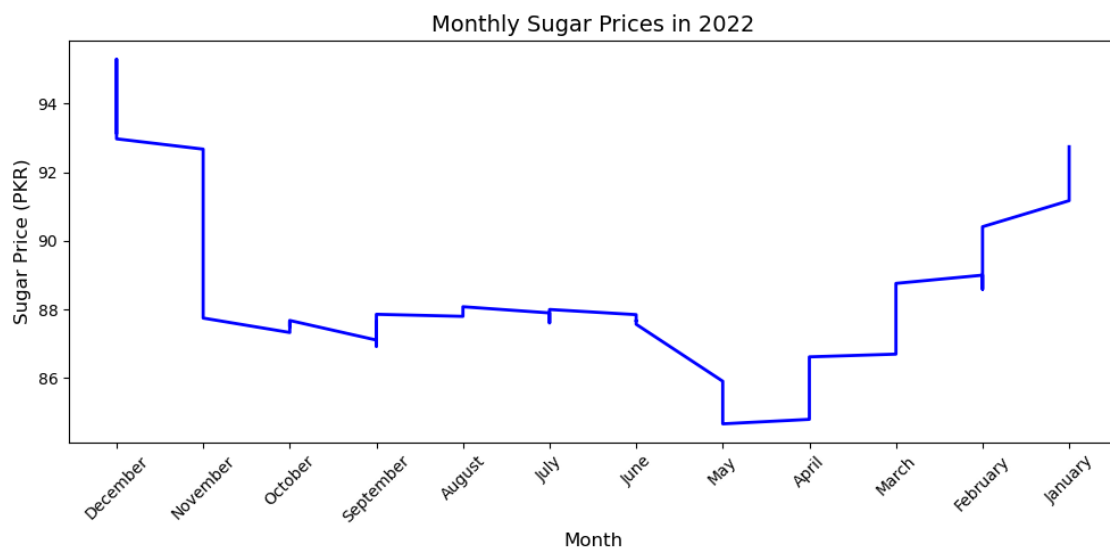
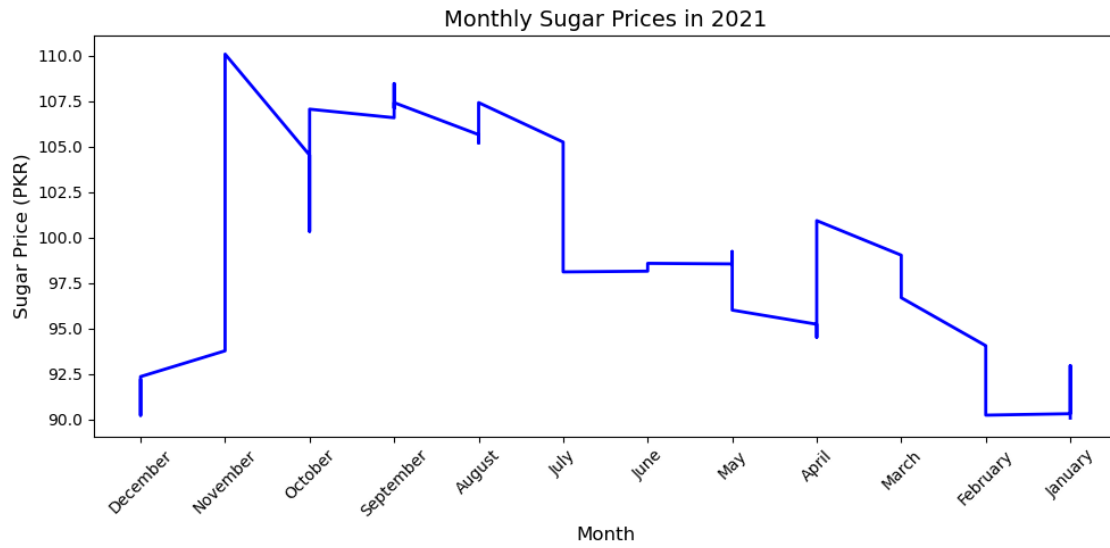
```

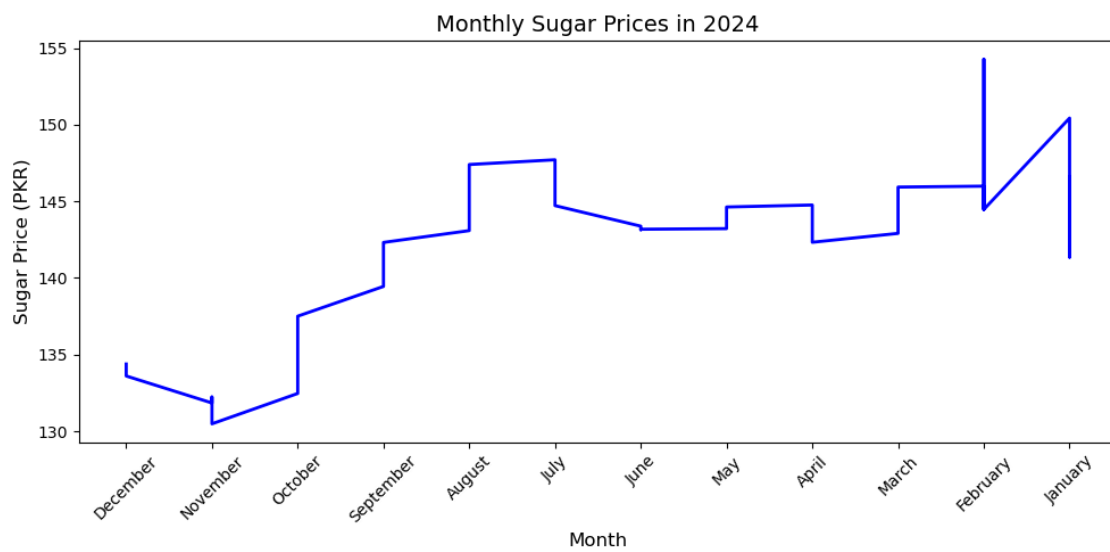
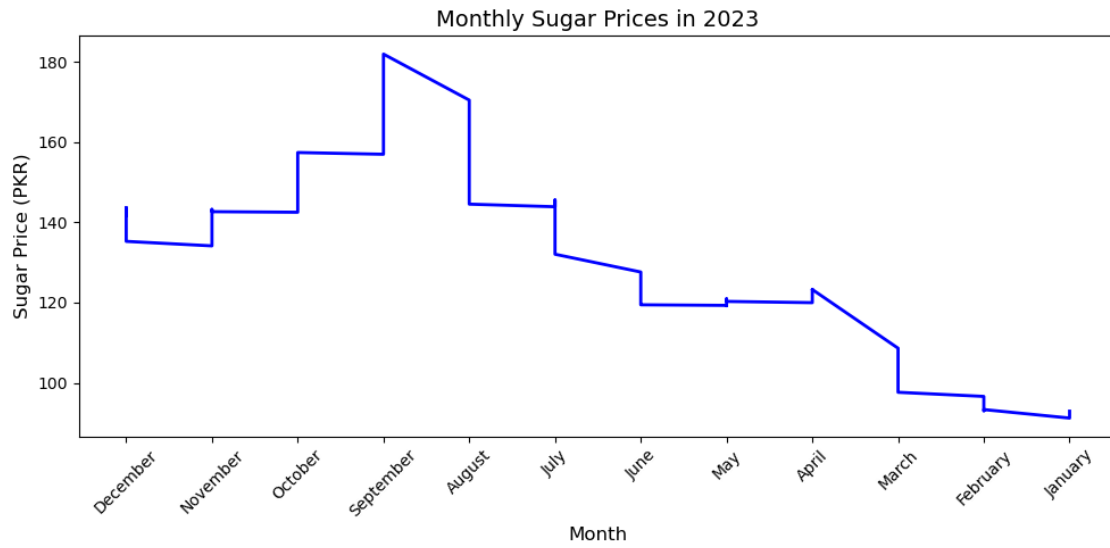






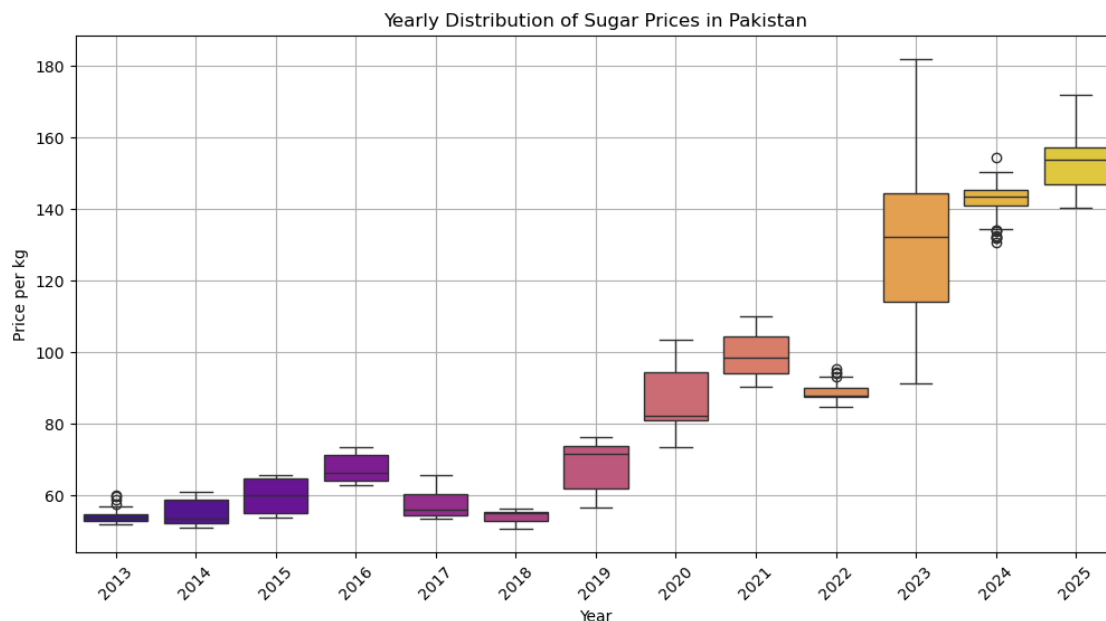






5.6 Yearly distribution of sugar prices in Pakistan along with outlier detection

[206] :



Prices were relatively stable from 2013 to 2018, as the IQRs are smaller.

Volatility starts increasing from 2019 onwards (with the exception fo 2022 and 2024), with 2023 showing the highest price fluctuation, suggesting market instability, supply chain disruptions, or policy impacts.

There are some points above the upper whisker, indicating that in 2022, there were some significantly high sugar prices compared to the general distribution of that year.

Unlike 2023, where there was a wider spread of prices, 2024 seems to have a more consistent price range. There were notable price drops (outliers below the lower whisker), which could indicate occasional downward price corrections. The majority of prices remain in a higher range, suggesting that sugar prices in 2024 were still elevated compared to previous years.

```
[49]: suger_prices_2013_2025_pak_cleaned.head()
```

```
[49]:   Period End  Islamabad  Rawalpindi  Gujranwala  Sialkot  Lahore  Faisalabad  \
1  2025-03-13    176.24    177.65    175.0    175.0    172.94    170.0
2  2025-03-07    169.41    169.66    165.0    165.0    163.32    160.0
3  2025-02-27    163.74    161.65    160.0    160.0    160.00    155.0
4  2025-02-20    161.24    159.66    155.0    158.0    156.41    155.0
5  2025-02-13    158.74    158.65    160.0    160.0    156.41    150.0

   Sargodha  Multan  Bahawalpur  ...  Sukkur  Larkana  Peshawar  Bannu  \
1    170.0    168.99    170.00  ...    168.0    168.32    171.10    170.0
2    160.0    158.99    160.00  ...    158.0    158.32    166.65    160.0
3    155.0    156.42    155.00  ...    155.0    153.32    160.55    155.0
4    155.0    154.20    153.32  ...    150.0    150.00    160.55    155.0
```

```
5      152.0  154.20      151.65  ...   150.0   150.00   156.65  150.0
```

```
      Quetta  Khuzdar  Pakistan  Year  Month  Month Name
1  174.49   170.67   171.90  2025     3      March
2  165.99   162.65   162.64  2025     3      March
3  158.99   157.65   157.68  2025     2  February
4  157.49   155.99   155.27  2025     2  February
5  154.99   155.99   154.27  2025     2  February
```

```
[5 rows x 22 columns]
```

```
[53]: yearly_suger_prices_2013_2025_pak_cleaned =   
      ↳suger_prices_2013_2025_pak_cleaned[['Period End', 'Year', 'Month', 'Month_  
      ↳Name', 'Pakistan' ]]  
yearly_suger_prices_2013_2025_pak_cleaned.head()
```

```
[53]:   Period End  Year  Month Month Name  Pakistan
1  2025-03-13  2025     3      March    171.90
2  2025-03-07  2025     3      March    162.64
3  2025-02-27  2025     2  February    157.68
4  2025-02-20  2025     2  February    155.27
5  2025-02-13  2025     2  February    154.27
```

```
[54]: yearly_suger_prices_2013_2025_pak_cleaned.dtypes
```

```
[54]: Period End    datetime64[ns]
Year              int32
Month             int32
Month Name        object
Pakistan          float64
dtype: object
```

```
[55]: yearly_suger_prices_2013_2025_pak_cleaned.describe()
```

```
[55]:
```

	Period End	Year	Month	Pakistan
count	574	574.000000	574.000000	574.000000
mean	2019-05-29 13:30:18.815331072	2018.918118	6.395470	83.273589
min	2013-01-03 00:00:00	2013.000000	1.000000	50.560000
25%	2016-08-05 18:00:00	2016.000000	3.000000	55.807500
50%	2019-08-12 00:00:00	2019.000000	6.500000	72.440000
75%	2022-05-31 06:00:00	2022.000000	9.000000	96.665000
max	2025-03-13 00:00:00	2025.000000	12.000000	181.980000
std	NaN	3.483482	3.497002	31.645498

5.6.1 Applying ARIMA

Checking for stationarity using ADF Test

```
[58]: # Importing the required libraries:
```

```
[59]: from statsmodels.tsa.stattools import adfuller
```

```
[60]: adf_result = adfuller(yearly_suger_prices_2013_2025_pak_cleaned['Pakistan'])
```

```
[61]: print('ADF Statistic:', adf_result[0])
      print('p-value:', adf_result[1])
```

ADF Statistic: -2.042512784585313

p-value: 0.26824924914209286

```
[62]: print('Critical Values:')
      for key, value in adf_result[4].items():
          print(f'    {key}: {value}')
```

Critical Values:

1%: -3.4419153151868147

5%: -2.8666418984730146

10%: -2.569487157434041

```
[63]: # if adf_result[1] <= 0.05:
      #     print("The time series is likely stationary.")
      # else:
      #     print("The time series is likely non-stationary.")

      if adf_result[1] <= 0.05:
          print("Strong evidence against the null hypothesis, reject the null_
          ↳hypothesis.The time series is likely stationary. ")
      else:
          print("Weak evidence against null hypothesis, time series is likely_
          ↳non-stationary .")
```

Weak evidence against null hypothesis, time series is likely non-stationary .

Applying differencing to make the time series stationary:

```
[65]: prices = yearly_suger_prices_2013_2025_pak_cleaned['Pakistan']
      period_end = yearly_suger_prices_2013_2025_pak_cleaned['Period End']

      prices_diff = prices.diff().dropna()
      period_end_for_diff = period_end[1:] # Corresponding 'Period End' values

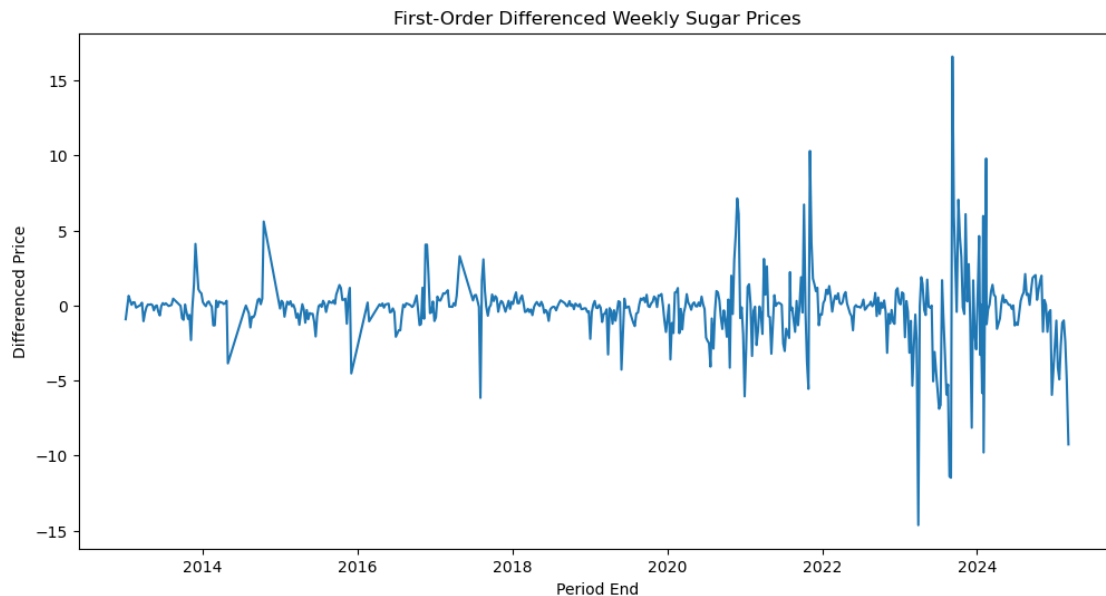
      # Verify the lengths
      print(f"Length of differenced prices: {len(prices_diff)}")
      print(f"Length of corresponding Period End: {len(period_end_for_diff)}")
```

Length of differenced prices: 573

Length of corresponding Period End: 573

```
[66]: # Plot the differenced series
      plt.figure(figsize=(12, 6))
```

```
plt.plot(period_end_for_diff, prices_diff)
plt.title('First-Order Differenced Weekly Sugar Prices')
plt.xlabel('Period End')
plt.ylabel('Differenced Price')
plt.show()
```



```
[67]: # Perform the ADF test on the differenced series
adf_result_diff = adfuller(prices_diff)
print('\nADF Statistic (Differenced):', adf_result_diff[0])
print('p-value (Differenced):', adf_result_diff[1])
print('Critical Values (Differenced):')
for key, value in adf_result_diff[4].items():
    print(f'    {key}: {value}')

if adf_result_diff[1] <= 0.05:
    print("The first-order differenced time series is likely stationary.")
else:
    print("The first-order differenced time series is likely non-stationary.")
```

ADF Statistic (Differenced): -9.621732761898885

p-value (Differenced): 1.7019084792975715e-16

Critical Values (Differenced):

1%: -3.4419153151868147

5%: -2.8666418984730146

10%: -2.569487157434041

The first-order differenced time series is likely stationary.

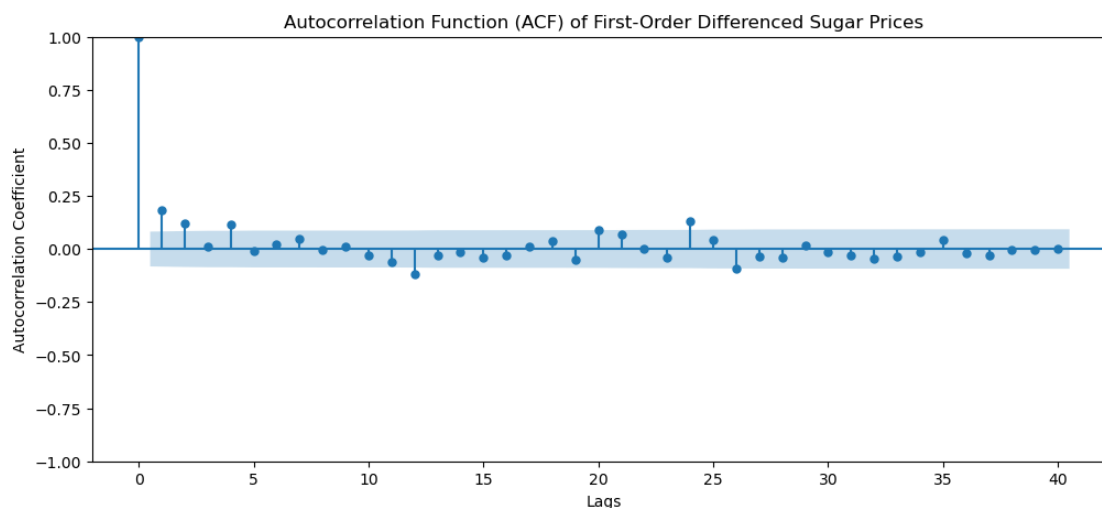
Determine the order of the Autoregressive (AR) component (p) and the Moving Average (MA) component (q).

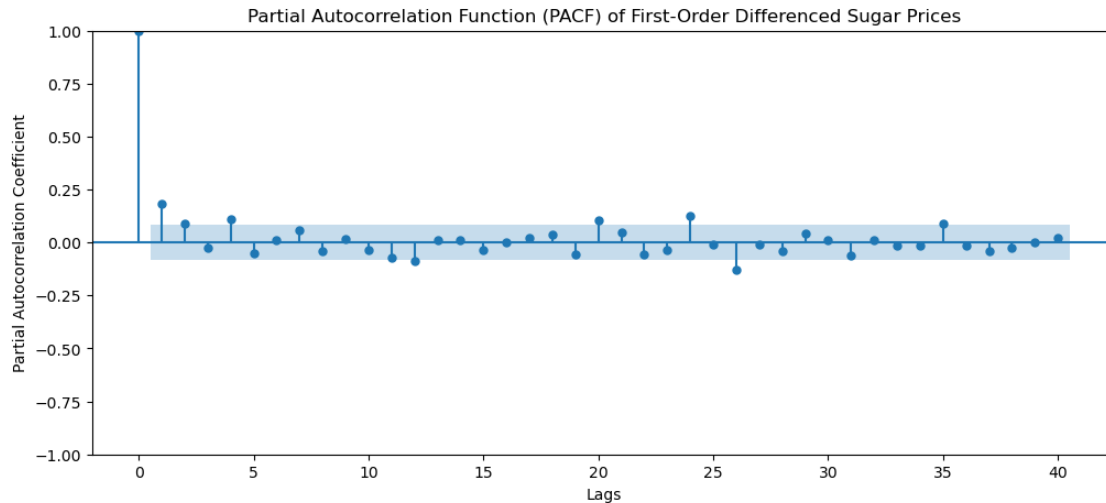
To do this, we will analyze the Autocorrelation Function (ACF) and the Partial Autocorrelation Function (PACF) plots of the first-order differenced stationary time series.

```
[70]: from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
```

```
[71]: # Plot ACF
plt.figure(figsize=(12, 5))
plot_acf(prices_diff, lags=40, ax=plt.gca()) # You can adjust the number of lags
plt.title('Autocorrelation Function (ACF) of First-Order Differenced Sugar_
↪Prices')
plt.xlabel('Lags')
plt.ylabel('Autocorrelation Coefficient')
plt.show()

# Plot PACF
plt.figure(figsize=(12, 5))
plot_pacf(prices_diff, lags=40, method='ywml', ax=plt.gca()) # You can adjust_
↪the number of lags
plt.title('Partial Autocorrelation Function (PACF) of First-Order Differenced_
↪Sugar Prices')
plt.xlabel('Lags')
plt.ylabel('Partial Autocorrelation Coefficient')
plt.show()
```





Fitting the ARIMA(1, 1, 1) Model to the sugar price data based on a rolling window approach

```
[75]: from statsmodels.tsa.arima.model import ARIMA
      from sklearn.metrics import mean_squared_error
      import warnings

[76]: # Assuming yearly_suger_prices_2013_2025_pak_cleaned is a pandas Series or a list
      data = yearly_suger_prices_2013_2025_pak_cleaned.dropna()
      window_size = int(0.7 * len(data))
      train_data = data[:window_size]

      # Store forecasts and actual values
      forecasted_prices = []
      actual_prices = []

      # The remaining 30% of the data will be our "future" for rolling prediction
      n_predictions = len(data) - window_size

      history = list(train_data['Pakistan'])

      for i in range(n_predictions):
          try:
              # Fit the ARIMA(1,1,1) model
              # warnings filter to avoid showing convergence warnings
              with warnings.catch_warnings():
                  warnings.filterwarnings("ignore")
                  model = ARIMA(history, order=(1, 1, 1))
                  model_fit = model.fit()
```



```

    # Make a one-step-ahead prediction
    output = model_fit.forecast()
    yhat = output[0]
    forecasted_prices.append(yhat)

    # Get the actual value for this time step
    obs = data>window_size + i]
    actual_prices.append(obs)

    # Add the predicted value to the history for the next iteration
    history.append(yhat)

    print(f"Iteration {i+1}: Forecast = {yhat:.2f}, Actual = {obs:.2f}") #  

    ↪Added for clarity

except Exception as e:
    print(f"Error at iteration {i+1}: {e}")
    # You might want to handle the error differently, e.g.,
    # - Use a simple prediction (like the last actual value)
    # - Skip this iteration
    # - Stop the loop
    break # I'll stop the loop for this example, but you can change it

print("Forecasted Prices:", forecasted_prices)
print("Actual Prices:", actual_prices)

```

Error at iteration 1: 401

Forecasted Prices: [60.09137935595068]

Actual Prices: []

ACF of Residuals:

This plot shows the autocorrelation of the residuals at different lags.

Ideally, we want the residuals to be uncorrelated (white noise).

While most of the autocorrelations are within the confidence intervals, there is a significant spike at lag 1. This indicates that there's some remaining autocorrelation in the residuals that the ARIMA(1, 1, 1) model hasn't fully captured. This suggests the model could be improved.

PACF of Residuals:

This plot shows the partial autocorrelation of the residuals. Similar to the ACF plot, there's a significant spike at lag 1, further confirming the presence of residual autocorrelation.

Histogram of Residuals:

This plot shows the distribution of the residuals.

The histogram is not bell-shaped, indicating that the residuals are not normally distributed. This confirms the result of the Jarque-Bera test in the model summary, which strongly rejected the null

hypothesis of normality. The distribution appears to be skewed to the left and has heavier tails than a normal distribution.

Summary of Residual Analysis (Based on Model Fitted to Original Data):

Heteroskedasticity: Evident in the time series plot and confirmed by the Heteroskedasticity test.

Autocorrelation: Significant autocorrelation at lag 1, indicating that the ARIMA(1, 1, 1) model has not fully accounted for the short-term dependencies in the data.

Non-Normality: The residuals are not normally distributed, as shown by the histogram and the Jarque-Bera test.

```
[79]: data = yearly_suger_prices_2013_2025_pak_cleaned.dropna()
data_sorted_year = data.sort_values(by='Period End', ascending=True).
      ↪reset_index().drop(columns=['index'])

window_size = int(0.7 * len(data))
n_predictions = len(data_sorted_year) - window_size
train_data = data_sorted_year[:window_size]
actual_prices = []
rolling_predictions_arima = []
Period_End=[]
train_prices = list(train_data['Pakistan'])
for i in range(n_predictions):
    train_window = train_prices[i:i + window_size]
    model = ARIMA(train_window, order=(1, 1, 1)) # Adjust the order as needed
    fitted_model = model.fit()
    forecast = fitted_model.forecast(steps=1)[0]
    train_prices.append(forecast)
    rolling_predictions_arima.append(forecast)
    actual = data_sorted_year['Pakistan'].iloc[window_size + i]
    period_end = data_sorted_year['Period End'].iloc[window_size + i]
    Period_End.append(period_end)
    actual_prices.append(actual)

rolling_predictions_arima = pd.Series(rolling_predictions_arima)
actual_prices = pd.Series(actual_prices)

# Calculate Mean Squared Error
mse_arima = mean_squared_error(actual_prices, rolling_predictions_arima)

# Create a DataFrame
forecast_df_ARIMA = pd.DataFrame({'Actual_Price': actual_prices,
      ↪'Forecasted_Price_ARIMA': rolling_predictions_arima})
forecast_df_ARIMA['MSE_ARIMA'] = mse_arima
forecast_df_ARIMA['Period End'] = Period_End
```

```

forecast_df_ARIMA = forecast_df_ARIMA[['Period End', 'Actual_Price', '
↳ 'Forecasted_Price_ARIMA', 'MSE_ARIMA']]
print("\nForecast DataFrame:")
forecast_df_ARIMA.set_index("Period End", inplace=True)
forecast_df_ARIMA.head(40)

```

Forecast DataFrame:

```

[79]:

```

	Actual_Price	Forecasted_Price_ARIMA	MSE_ARIMA
Period End			
2021-11-04	110.08	105.634345	932.662572
2021-11-11	99.79	106.110813	932.662572
2021-11-18	95.55	106.314871	932.662572
2021-11-25	93.76	106.402187	932.662572
2021-12-02	92.34	106.439535	932.662572
2021-12-09	91.39	106.455513	932.662572
2021-12-16	90.21	106.462350	932.662572
2021-12-23	91.53	106.465275	932.662572
2021-12-30	92.12	106.466527	932.662572
2022-01-06	92.74	106.467062	932.662572
2022-01-13	92.58	106.467291	932.662572
2022-01-20	92.22	106.467390	932.662572
2022-01-27	91.17	106.467432	932.662572
2022-02-03	90.41	106.467450	932.662572
2022-02-10	89.12	106.467457	932.662572
2022-02-17	88.59	106.467461	932.662572
2022-02-24	89.00	106.467462	932.662572
2022-03-03	88.76	106.467463	932.662572
2022-03-10	88.11	106.467463	932.662572
2022-03-17	87.68	106.467463	932.662572
2022-03-24	86.87	106.467463	932.662572
2022-03-31	86.70	106.467463	932.662572
2022-04-07	86.62	106.467463	932.662572
2022-04-14	86.40	106.467463	932.662572
2022-04-21	85.69	106.467463	932.662572
2022-04-28	84.80	106.467463	932.662572
2022-05-12	84.67	106.467463	932.662572
2022-05-19	85.20	106.467463	932.662572
2022-05-26	85.91	106.467463	932.662572
2022-06-02	87.57	106.467463	932.662572
2022-06-09	87.68	106.467463	932.662572
2022-06-16	87.66	106.467463	932.662572
2022-06-23	87.76	106.467463	932.662572
2022-06-30	87.85	106.467463	932.662572
2022-07-06	88.00	106.467463	932.662572
2022-07-14	88.00	106.467463	932.662572

2022-07-21	87.61	106.467463	932.662572
2022-07-28	87.90	106.467463	932.662572
2022-08-04	88.08	106.467463	932.662572
2022-08-11	88.08	106.467463	932.662572

The results shown by ARIMA (1,1,1) are not accurate . We will focus on auto ARIMA for forecasting purposes

```
[55]: pip install pmdarima
```

```
[65]: data_sorted_year = data.sort_values(by='Period End', ascending=True).
      ↪reset_index(drop=True)
window_size = int(0.7 * len(data_sorted_year))
n_predictions = len(data_sorted_year) - window_size
print(f"Window Size: {window_size}")
print(f"Number of predictions: {n_predictions}")
# took 5.06 minutes to run 150 iterations
# 2.69 mins to return 100 predictions
# 7.89 minutes to run 150
```

Window Size: 401

Number of predictions: 173

```
[150]: # Finaal code Sun 11th May

# auto arima with sliding window using actual prices
# correct code with auto arima
from pmdarima.arima import auto_arima
from sklearn.metrics import mean_squared_error
import time # Import the time module

# Prepare data
data = yearly_suger_prices_2013_2025_pak_cleaned.dropna()
data_sorted_year = data.sort_values(by='Period End', ascending=True).
      ↪reset_index(drop=True)

window_size = int(0.7 * len(data_sorted_year))
n_predictions = len(data_sorted_year) - window_size

train_window = list(data_sorted_year['Pakistan'].iloc[:window_size])
rolling_predictions_auto_arima = []
actual_prices = []
Period_End = []

start_time = time.time() # Record the start time

for i in range(n_predictions):
```

```

    # Auto ARIMA on current window (disable stepwise for better accuracy but
    ↪ slower)
    model = auto_arma(train_window,
                      seasonal=False,
                      suppress_warnings=True,
                      error_action='ignore',
                      stepwise=True) # or stepwise=False for exhaustive search

    # Forecast 1 step ahead
    forecast = model.predict(n_periods=1)[0]

    # Store forecast and actual
    actual = data_sorted_year['Pakistan'].iloc>window_size + i]
    rolling_predictions_auto_arma.append(forecast)
    actual_prices.append(actual)
    Period_End.append(data_sorted_year['Period End'].iloc>window_size + i])

    # Sliding window update: drop oldest, add forecast
    train_window.pop(0)
    train_window.append(actual)

end_time = time.time() # Record the end time
execution_time = end_time - start_time # Calculate execution time
execution_time_min = execution_time / 60
# Final forecast evaluation
forecast_df_auto_arma = pd.DataFrame({
    'Period End': Period_End,
    'Actual_Price': actual_prices,
    'Forecasted_Price_AutoARIMA': rolling_predictions_auto_arma
})
forecast_df_auto_arma['MSE_AutoARIMA'] = mean_squared_error(
    forecast_df_auto_arma['Actual_Price'],
    forecast_df_auto_arma['Forecasted_Price_AutoARIMA']
)

forecast_df_auto_arma.set_index("Period End", inplace=True)
print(forecast_df_auto_arma.head(40))
print(f"Execution Time: {execution_time_min:.2f} minutes") # Print the
↪ execution time

```

	Actual_Price	Forecasted_Price_AutoARIMA	MSE_AutoARIMA
Period End			
2021-11-04	110.08	106.104164	12.706436
2021-11-11	99.79	112.918010	12.706436
2021-11-18	95.55	97.318798	12.706436
2021-11-25	93.76	93.424716	12.706436
2021-12-02	92.34	94.632116	12.706436
2021-12-09	91.39	91.955558	12.706436

2021-12-16	90.21	91.131990	12.706436
2021-12-23	91.53	89.888334	12.706436
2021-12-30	92.12	91.886994	12.706436
2022-01-06	92.74	92.279766	12.706436
2022-01-13	92.58	92.908078	12.706436
2022-01-20	92.22	92.536575	12.706436
2022-01-27	91.17	92.122452	12.706436
2022-02-03	90.41	90.885145	12.706436
2022-02-10	89.12	90.203395	12.706436
2022-02-17	88.59	88.768149	12.706436
2022-02-24	89.00	88.445308	12.706436
2022-03-03	88.76	89.111798	12.706436
2022-03-10	88.11	88.694586	12.706436
2022-03-17	87.68	87.932747	12.706436
2022-03-24	86.87	87.562663	12.706436
2022-03-31	86.70	86.648660	12.706436
2022-04-07	86.62	86.653585	12.706436
2022-04-14	86.40	86.598159	12.706436
2022-04-21	85.69	86.339931	12.706436
2022-04-28	84.80	85.496036	12.706436
2022-05-12	84.67	84.556381	12.706436
2022-05-19	85.20	84.634430	12.706436
2022-05-26	85.91	85.344973	12.706436
2022-06-02	87.57	86.104443	12.706436
2022-06-09	87.68	88.026580	12.706436
2022-06-16	87.66	87.710188	12.706436
2022-06-23	87.76	87.654508	12.706436
2022-06-30	87.85	87.787400	12.706436
2022-07-06	88.00	87.874666	12.706436
2022-07-14	88.00	88.041118	12.706436
2022-07-21	87.61	88.000000	12.706436
2022-07-28	87.90	87.503297	12.706436
2022-08-04	88.08	87.979293	12.706436
2022-08-11	88.08	88.129126	12.706436

Execution Time: 6.35 minutes

```
[152]: forecast_df_auto_arima.tail(40)
```

```
[152]:
```

	Actual_Price	Forecasted_Price_AutoARIMA	MSE_AutoARIMA
Period End			
2024-05-30	143.22	143.572878	12.706436
2024-06-06	143.18	143.294124	12.706436
2024-06-13	143.15	143.409719	12.706436
2024-06-20	143.38	143.303634	12.706436
2024-06-27	143.38	143.565809	12.706436
2024-07-04	144.72	143.589746	12.706436
2024-07-11	145.88	145.050178	12.706436

2024-07-18	147.18	146.626199	12.706436
2024-07-25	147.71	147.213845	12.706436
2024-08-01	147.40	148.195311	12.706436
2024-08-08	146.75	147.808020	12.706436
2024-08-15	145.88	146.461390	12.706436
2024-08-22	143.79	145.796794	12.706436
2024-08-29	143.09	143.436348	12.706436
2024-09-05	142.32	142.971171	12.706436
2024-09-12	142.28	142.305672	12.706436
2024-09-19	141.28	141.784497	12.706436
2024-09-26	139.45	141.201602	12.706436
2024-10-03	137.51	139.017303	12.706436
2024-10-10	135.49	137.161767	12.706436
2024-10-17	135.12	135.108169	12.706436
2024-10-24	134.08	135.005787	12.706436
2024-10-31	132.47	134.000125	12.706436
2024-11-07	130.49	132.057842	12.706436
2024-11-14	132.24	130.150502	12.706436
2024-11-21	131.88	132.562702	12.706436
2024-11-28	131.85	132.088406	12.706436
2024-12-05	133.61	131.548681	12.706436
2024-12-12	134.08	134.091955	12.706436
2024-12-19	134.39	134.198455	12.706436
2025-01-09	140.35	134.340059	12.706436
2025-01-16	141.36	141.401815	12.706436
2025-01-23	145.50	141.925050	12.706436
2025-01-30	150.43	145.676699	12.706436
2025-02-06	153.11	151.992870	12.706436
2025-02-13	154.27	153.579963	12.706436
2025-02-20	155.27	154.677551	12.706436
2025-02-27	157.68	155.623883	12.706436
2025-03-07	162.64	158.303185	12.706436
2025-03-13	171.90	163.809471	12.706436

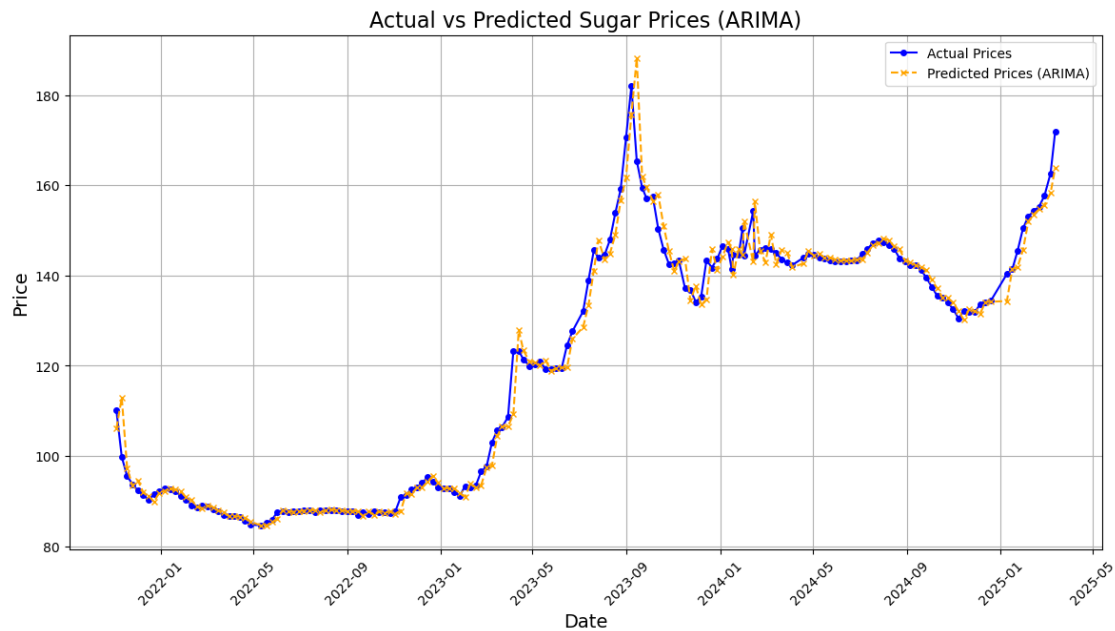
```
[154]: import matplotlib.pyplot as plt
```

```
# Plot the actual and predicted prices for ARIMA
plt.figure(figsize=(14, 7))
plt.plot(forecast_df_auto_arima.index,
         ↪forecast_df_auto_arima['Actual_Price'],label='Actual Prices', color='blue',
         ↪marker='o', markersize=4)
plt.plot(forecast_df_auto_arima.index,
         ↪forecast_df_auto_arima['Forecasted_Price_AutoARIMA'], label='Predicted Prices_
         ↪(ARIMA)', color='orange', linestyle='--', marker='x', markersize=4)

# Adding title and labels
plt.title('Actual vs Predicted Sugar Prices (ARIMA)', fontsize=16)
```

```
plt.xlabel('Date', fontsize=14)
plt.ylabel('Price', fontsize=14)
plt.xticks(rotation=45)
plt.legend()
plt.grid(True)

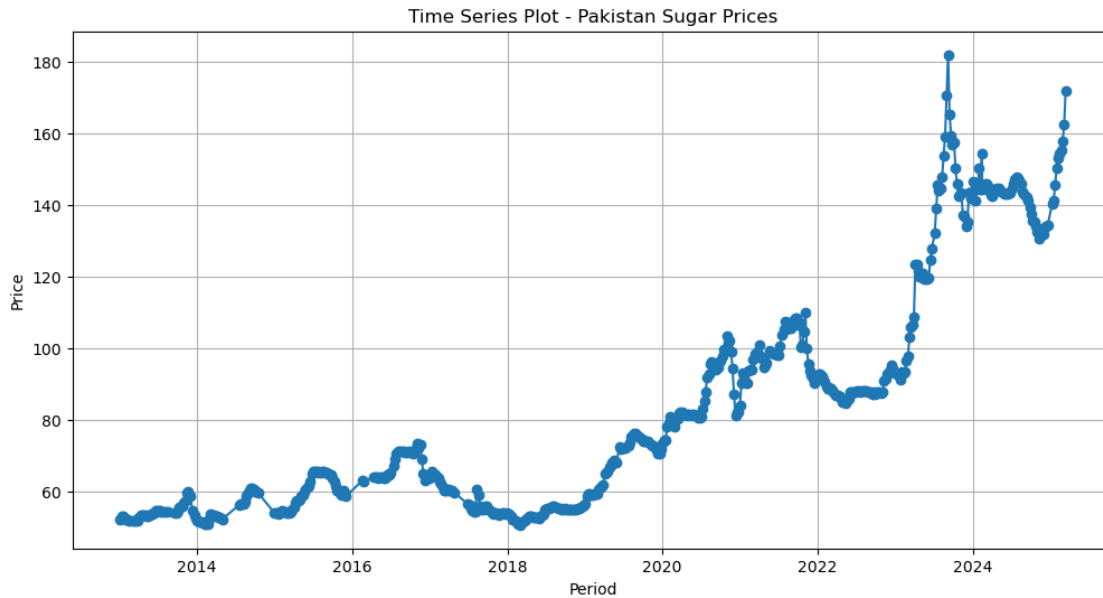
# Show plot
plt.show()
```



5.7 Exponential Smoothing:

```
[174]: import matplotlib.pyplot as plt

plt.figure(figsize=(12, 6))
plt.plot(data_sorted_year['Period End'], data_sorted_year['Pakistan'],
         ↪marker='o')
plt.title('Time Series Plot - Pakistan Sugar Prices')
plt.xlabel('Period')
plt.ylabel('Price')
plt.grid(True)
plt.show()
```

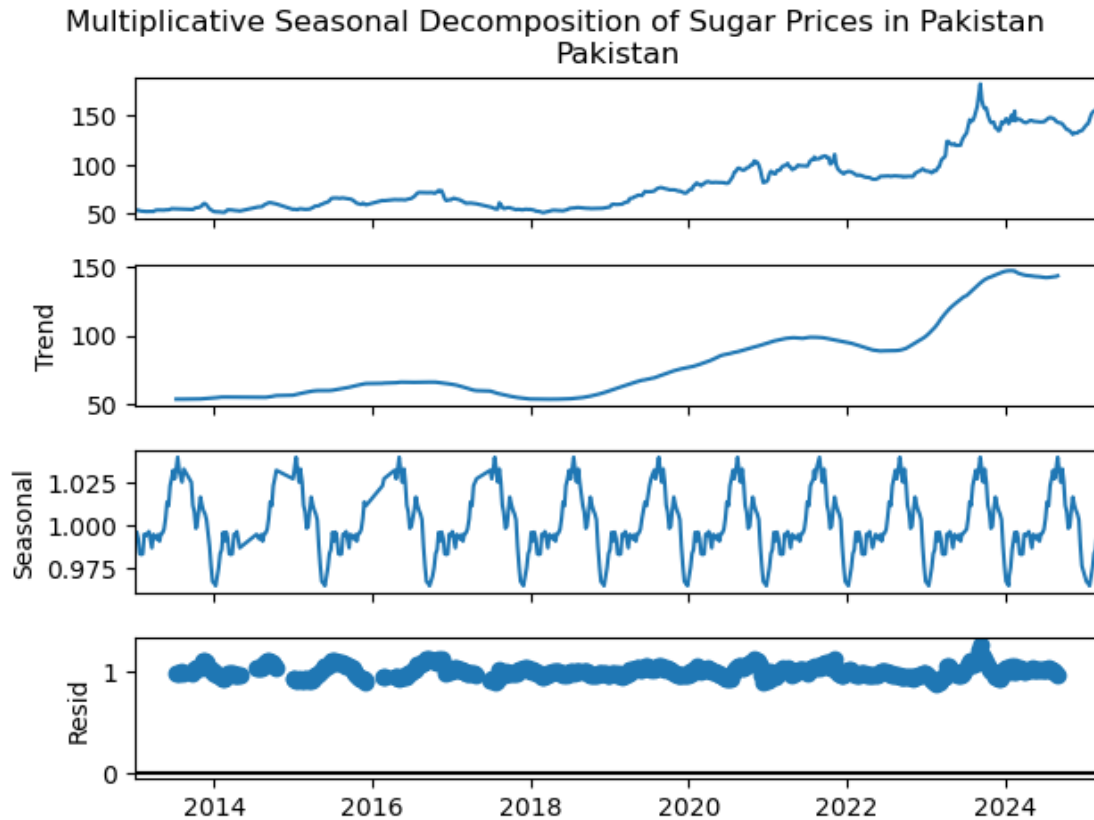



```
[202]: from statsmodels.tsa.seasonal import seasonal_decompose
import matplotlib.pyplot as plt

# Prepare the time series
ts = data_sorted_year.set_index('Period End')['Pakistan']

# Perform additive seasonal decomposition
decomposition = seasonal_decompose(ts, model='multiplicative', period=52) #
    ↳ Adjust period based on actual seasonality

# Plot the decomposition with a custom title
fig = decomposition.plot()
fig.suptitle('Multiplicative Seasonal Decomposition of Sugar Prices in
    ↳ Pakistan', fontsize=12)
plt.tight_layout()
plt.subplots_adjust(top=0.9) # Adjust top spacing to fit title
plt.show()
```



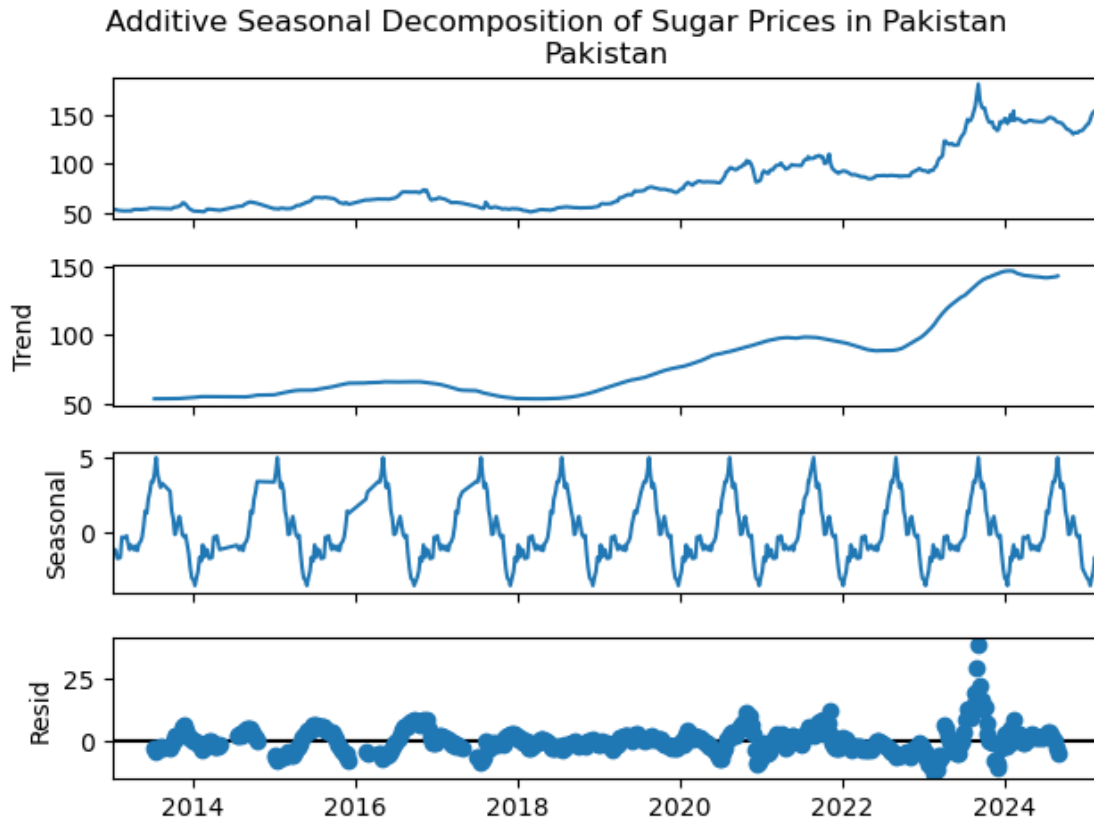
The seasonal component clearly shows a repeating pattern that occurs approximately every year

```
[198]: # import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import seasonal_decompose

# Prepare the time series
ts = data_sorted_year.set_index('Period End')['Pakistan']

# Perform additive seasonal decomposition
decomposition = seasonal_decompose(ts, model='additive', period=52) # Adjust_
    ↳ period based on actual seasonality

# Plot the decomposition with a custom title
fig = decomposition.plot()
fig.suptitle('Additive Seasonal Decomposition of Sugar Prices in Pakistan',
    ↳ fontsize=12)
plt.tight_layout()
plt.subplots_adjust(top=0.9) # Adjust top spacing to fit title
plt.show()
```



```
[73]: from statsmodels.tsa.seasonal import seasonal_decompose
import matplotlib.pyplot as plt

ts = data_sorted_year.set_index('Period End')['Pakistan']

additive_result = seasonal_decompose(ts, model='additive', period=52)
multiplicative_result = seasonal_decompose(ts, model='multiplicative', period=52)

# Calculate standard deviation of residuals
additive_resid_std = additive_result.resid.dropna().std()
multiplicative_resid_std = multiplicative_result.resid.dropna().std()

print(f"Additive Residual Std Dev: {additive_resid_std:.4f}")
print(f"Multiplicative Residual Std Dev: {multiplicative_resid_std:.4f}")
```

Additive Residual Std Dev: 4.9336

Multiplicative Residual Std Dev: 0.0516

If the multiplicative residual standard deviation is much lower than the additive residual standard deviation, it suggests that the multiplicative model is capturing the variability of the data more effectively in relative terms (i.e., percentage errors are smaller).

Multiplicative models are often more suitable for data where the magnitude of the trend or seasonality changes over time in proportion to the level of the series (i.e., the size of fluctuations grows or shrinks with the level of the series).

```
[222]: import pandas as pd
from statsmodels.tsa.holtwinters import ExponentialSmoothing
from sklearn.metrics import mean_squared_error
import time

# Prepare data
data = yearly_suger_prices_2013_2025_pak_cleaned.dropna()
data_sorted_year = data.sort_values(by='Period End', ascending=True).
    ↪reset_index(drop=True)

window_size = int(0.7 * len(data_sorted_year))
n_predictions = len(data_sorted_year) - window_size

train_window = list(data_sorted_year['Pakistan'].iloc[:window_size])
rolling_predictions_ets = []
actual_prices = []
Period_End = []

start_time = time.time() # Start timer

for i in range(n_predictions):
    # Fit Exponential Smoothing (no trend/seasonality by default; change if
    ↪needed)
    model = ExponentialSmoothing(train_window, trend='mul', seasonal='mul',
    ↪seasonal_periods=12)
    model_fit = model.fit(optimized=True)

    # Forecast 1 step ahead
    forecast = model_fit.forecast(1)[0]

    # Store forecast and actual
    actual = data_sorted_year['Pakistan'].iloc[window_size + i]
    rolling_predictions_ets.append(forecast)
    actual_prices.append(actual)
    Period_End.append(data_sorted_year['Period End'].iloc[window_size + i])

    # Update sliding window
    train_window.pop(0)
    train_window.append(actual)

end_time = time.time()
execution_time_min = (end_time - start_time) / 60
```

```

# Final forecast evaluation
forecast_df_ets = pd.DataFrame({
    'Period_End': Period_End,
    'Actual_Price': actual_prices,
    'Forecasted_Price_ETS': rolling_predictions_ets
})
forecast_df_ets['MSE_ETS'] = mean_squared_error(
    forecast_df_ets['Actual_Price'],
    forecast_df_ets['Forecasted_Price_ETS']
)

forecast_df_ets.set_index("Period_End", inplace=True)
print(forecast_df_ets.head(40))
print(f"Execution Time: {execution_time_min:.2f} minutes")

```

	Actual_Price	Forecasted_Price_ETS	MSE_ETS
Period_End			
2021-11-04	110.08	104.120319	13.105844
2021-11-11	99.79	110.545305	13.105844
2021-11-18	95.55	99.548719	13.105844
2021-11-25	93.76	95.671499	13.105844
2021-12-02	92.34	94.047443	13.105844
2021-12-09	91.39	92.900743	13.105844
2021-12-16	90.21	91.814346	13.105844
2021-12-23	91.53	90.739143	13.105844
2021-12-30	92.12	91.893121	13.105844
2022-01-06	92.74	91.724680	13.105844
2022-01-13	92.58	92.635211	13.105844
2022-01-20	92.22	93.167147	13.105844
2022-01-27	91.17	92.216325	13.105844
2022-02-03	90.41	90.903727	13.105844
2022-02-10	89.12	89.970488	13.105844
2022-02-17	88.59	89.129298	13.105844
2022-02-24	89.00	88.777016	13.105844
2022-03-03	88.76	89.473280	13.105844
2022-03-10	88.11	89.104197	13.105844
2022-03-17	87.68	88.644301	13.105844
2022-03-24	86.87	88.031625	13.105844
2022-03-31	86.70	86.536354	13.105844
2022-04-07	86.62	86.599398	13.105844
2022-04-14	86.40	87.133976	13.105844
2022-04-21	85.69	86.349122	13.105844
2022-04-28	84.80	85.413032	13.105844
2022-05-12	84.67	84.332973	13.105844
2022-05-19	85.20	84.653006	13.105844
2022-05-26	85.91	85.393026	13.105844
2022-06-02	87.57	86.329998	13.105844
2022-06-09	87.68	87.850467	13.105844

2022-06-16	87.66	88.175713	13.105844
2022-06-23	87.76	87.968993	13.105844
2022-06-30	87.85	87.425834	13.105844
2022-07-06	88.00	87.751088	13.105844
2022-07-14	88.00	88.500066	13.105844
2022-07-21	87.61	87.917172	13.105844
2022-07-28	87.90	87.296236	13.105844
2022-08-04	88.08	87.434780	13.105844
2022-08-11	88.08	88.085263	13.105844

Execution Time: 0.53 minutes

[224]: `forecast_df_ets.tail(40)`

[224]:

Period End	Actual_Price	Forecasted_Price_ETS	MSE_ETS
2024-05-30	143.22	143.780201	13.105844
2024-06-06	143.18	143.332483	13.105844
2024-06-13	143.15	143.049365	13.105844
2024-06-20	143.38	142.882747	13.105844
2024-06-27	143.38	143.279914	13.105844
2024-07-04	144.72	144.291632	13.105844
2024-07-11	145.88	145.911347	13.105844
2024-07-18	147.18	146.736670	13.105844
2024-07-25	147.71	147.685283	13.105844
2024-08-01	147.40	149.011182	13.105844
2024-08-08	146.75	145.918799	13.105844
2024-08-15	145.88	147.242404	13.105844
2024-08-22	143.79	146.234501	13.105844
2024-08-29	143.09	143.878915	13.105844
2024-09-05	142.32	142.896276	13.105844
2024-09-12	142.28	142.030140	13.105844
2024-09-19	141.28	142.128408	13.105844
2024-09-26	139.45	142.200489	13.105844
2024-10-03	137.51	140.617930	13.105844
2024-10-10	135.49	138.358416	13.105844
2024-10-17	135.12	135.969522	13.105844
2024-10-24	134.08	136.225391	13.105844
2024-10-31	132.47	132.766584	13.105844
2024-11-07	130.49	132.831706	13.105844
2024-11-14	132.24	130.652715	13.105844
2024-11-21	131.88	132.273132	13.105844
2024-11-28	131.85	131.684104	13.105844
2024-12-05	133.61	131.583490	13.105844
2024-12-12	134.08	133.369359	13.105844
2024-12-19	134.39	134.752488	13.105844
2025-01-09	140.35	135.379776	13.105844
2025-01-16	141.36	141.018798	13.105844

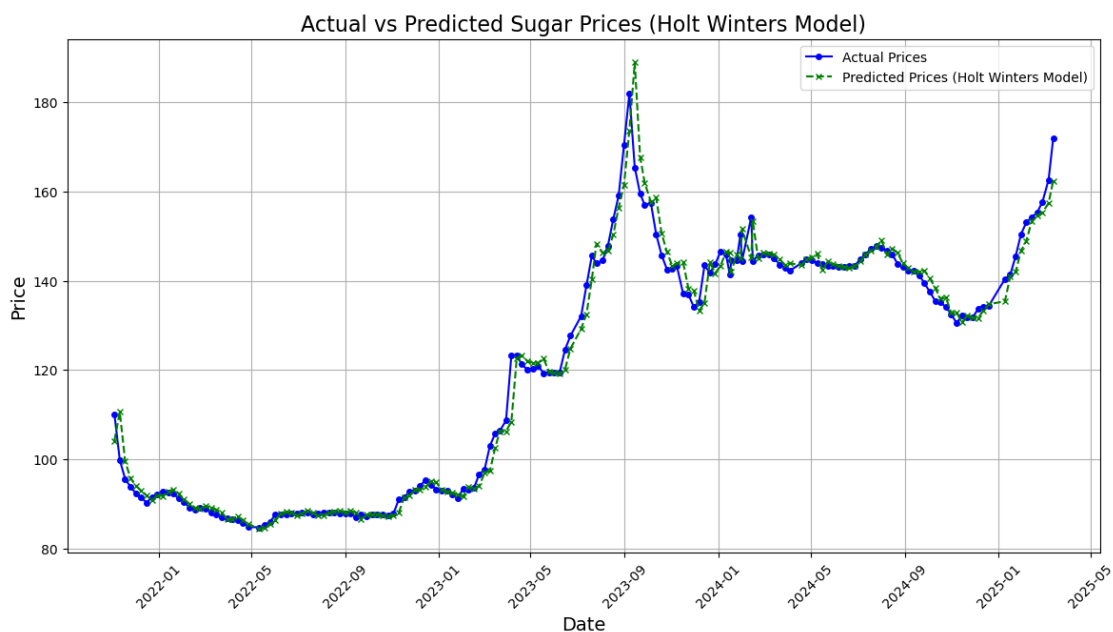
2025-01-23	145.50	141.952072	13.105844
2025-01-30	150.43	146.703385	13.105844
2025-02-06	153.11	148.980209	13.105844
2025-02-13	154.27	153.363849	13.105844
2025-02-20	155.27	154.553086	13.105844
2025-02-27	157.68	155.291781	13.105844
2025-03-07	162.64	157.418810	13.105844
2025-03-13	171.90	162.419654	13.105844

```
[226]: import matplotlib.pyplot as plt

# Plot the actual and predicted prices for Holt Winters model
plt.figure(figsize=(14, 7))
plt.plot(forecast_df_ets.index, forecast_df_ets['Actual_Price'], label='Actual_
↪Prices', color='blue', marker='o', markersize=4)
plt.plot(forecast_df_ets.index, forecast_df_ets['Forecasted_Price_ETS'],
↪label='Predicted Prices (Holt Winters Model)', color='green', linestyle='--',
↪marker='x', markersize=4)

# Adding title and labels
plt.title('Actual vs Predicted Sugar Prices (Holt Winters Model)', fontsize=16)
plt.xlabel('Date', fontsize=14)
plt.ylabel('Price', fontsize=14)
plt.xticks(rotation=45)
plt.legend()
plt.grid(True)

# Show plot
plt.show()
```



6 Random Forest Regression

7 70% train, 30% test split

```
[81]: data_sorted_year.head()
```

```
[81]:
```

	Period End	Year	Month	Month Name	Pakistan
0	2013-01-03	2013	1	January	52.00
1	2013-01-10	2013	1	January	52.92
2	2013-01-17	2013	1	January	53.09
3	2013-01-24	2013	1	January	52.45
4	2013-01-31	2013	1	January	52.13

```
[83]: suger_prices_2013_2025_pak.head()
```

```
[83]:
```

	Period End	Islamabad	Rawalpindi	Gujranwala	Sialkot	Lahore	\
0	Weekly	PKR	PKR	PKR	PKR	PKR	
1	13-Mar-2025	176.240000	177.650000	175	175	172.940000	
2	07-Mar-2025	169.410000	169.660000	165	165	163.320000	
3	27-Feb-2025	163.740000	161.650000	160	160	160	
4	20-Feb-2025	161.240000	159.660000	155	158	156.410000	

	Faisalabad	Sargodha	Multan	Bahawalpur	Karachi	Hyderabad	Sukkur	\
0	PKR	PKR	PKR	PKR	PKR	PKR	PKR	
1	170	170	168.990000	170	174.370000	170	168	
2	160	160	158.990000	160	162.540000	160	158	
3	155	155	156.420000	155	157.100000	156.650000	155	
4	155	155	154.200000	153.320000	153.280000	150	150	

	Larkana	Peshawar	Bannu	Quetta	Khuzdar	Pakistan
0	PKR	PKR	PKR	PKR	PKR	PKR
1	168.320000	171.100000	170	174.490000	170.670000	171.900000
2	158.320000	166.650000	160	165.990000	162.650000	162.640000
3	153.320000	160.550000	155	158.990000	157.650000	157.680000
4	150	160.550000	155	157.490000	155.990000	155.270000

```
[93]: # Prepare data

df = data_sorted_year

target_col = 'Pakistan'
df[target_col] = pd.to_numeric(df[target_col], errors='coerce')
df = df[['Period End', target_col]].dropna()
```



```
[95]: def create_lag_features(df, target_col, lags=12):
    for lag in range(1, lags+1):
        df[f'lag_{lag}'] = df[target_col].shift(lag)
    return df

lags = 12 # Use 12 previous periods
df_lagged = create_lag_features(df.copy(), target_col, lags)
df_lagged = df_lagged.dropna().reset_index(drop=True)

[123]: import pandas as pd
import numpy as np
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error, mean_absolute_error
import matplotlib.pyplot as plt

window_size = int(0.7 * len(df_lagged))
train_data = df_lagged[:window_size]
test_data = df_lagged[window_size:]

history = list(train_data[target_col])
forecasted_prices = []
actual_prices = []
forecast_dates = []

for i in range(len(test_data)):
    # Prepare lags for the current prediction
    lags_input = history[-12:] # 12 lags
    X_pred = np.array(lags_input).reshape(1, -1)

    # Train model on all available history up to this point
    train_window = df_lagged.iloc[:window_size + i]
    X_train = train_window.drop(['Period End', target_col], axis=1)
    y_train = train_window[target_col]
    model = RandomForestRegressor(n_estimators=100, random_state=42)
    model.fit(X_train, y_train)

    # Predict
    y_pred = model.predict(X_pred)[0]
    forecasted_prices.append(y_pred)
    actual_prices.append(test_data[target_col].iloc[i])
    forecast_dates.append(test_data['Period End'].iloc[i])

    # Update history with actual value (for true rolling forecast)
    history.append(test_data[target_col].iloc[i])

# Convert results to DataFrame
results = pd.DataFrame({
```

```

    'Date': forecast_dates,
    'Actual': actual_prices,
    'Predicted': forecasted_prices
})

```

```

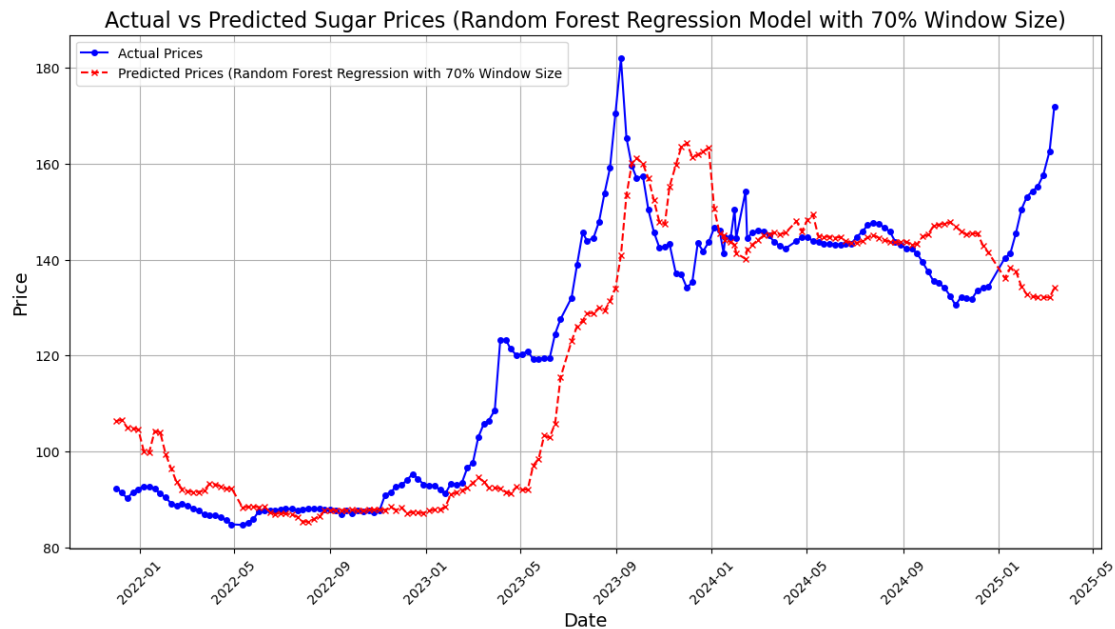
[125]: import matplotlib.pyplot as plt

# Plot the actual and predicted prices for Holt Winters model
plt.figure(figsize=(14, 7))
plt.plot(results['Date'], results['Actual'], label='Actual Prices',
         color='blue', marker='o', markersize=4)
plt.plot(results['Date'], results['Predicted'], label='Predicted Prices (Random
         Forest Regression with 70% Window Size', color='red', linestyle='--',
         marker='x', markersize=4)

# Adding title and labels
plt.title('Actual vs Predicted Sugar Prices (Random Forest Regression Model with
         70% Window Size)', fontsize=16)
plt.xlabel('Date', fontsize=14)
plt.ylabel('Price', fontsize=14)
plt.xticks(rotation=45)
plt.legend()
plt.grid(True)

# Show plot
plt.show()

```



```
[107]: # Evaluation
rmse = np.sqrt(mean_squared_error(results['Actual'], results['Predicted']))
mae = mean_absolute_error(results['Actual'], results['Predicted'])
print(f'RMSE: {rmse:.2f}')
print(f'MAE: {mae:.2f}')
```

RMSE: 12.75

MAE: 8.83

#Window Size 100

```
[113]: #Window Size 100

window_size = 100 # Number of samples in each training window
horizon = 1       # Forecast horizon (one step ahead)

preds_100 = []
actuals_100 = []
dates_100 = []

for i in range(window_size, len(df_lagged)):
    train = df_lagged.iloc[i-window_size:i]
    test = df_lagged.iloc[i:i+horizon]
    X_train = train.drop(['Period End', target_col], axis=1)
    y_train = train[target_col]
    X_test = test.drop(['Period End', target_col], axis=1)
    y_test = test[target_col]

    model = RandomForestRegressor(n_estimators=100, random_state=42)
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    preds_100.extend(y_pred)
    actuals_100.extend(y_test.values)
    dates_100.extend(test['Period End'].values)

# Convert results to DataFrame
results_100 = pd.DataFrame({'Date': dates, 'Actual': actuals, 'Predicted':
    ↪preds})
results_100['Date'] = pd.to_datetime(results['Date'])
```

```
[127]: import matplotlib.pyplot as plt

# Plot the actual and predicted prices for Holt Winters model
plt.figure(figsize=(14, 7))
plt.plot(results_100['Date'], results_100['Actual'], label='Actual Prices',
    ↪color='blue', marker='o', markersize=4)
```

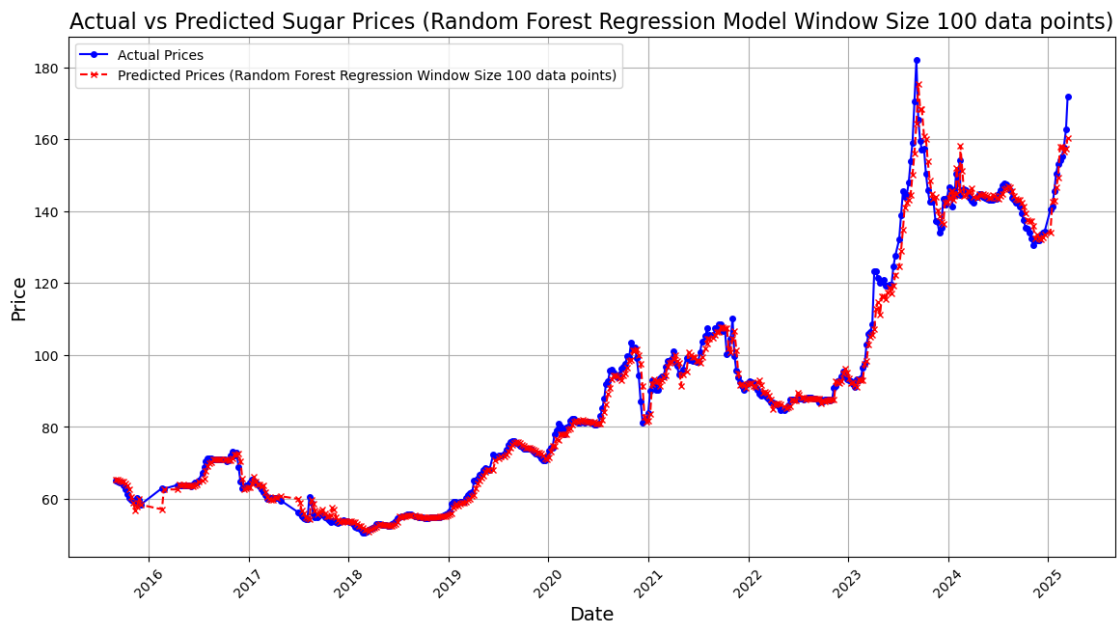
```

plt.plot(results_100['Date'], results_100['Predicted'], label='Predicted Prices,
↳(Random Forest Regression Window Size 100 data points)', color='red',
↳linestyle='--', marker='x', markersize=4)

# Adding title and labels
plt.title('Actual vs Predicted Sugar Prices (Random Forest Regression Model,
↳Window Size 100 data points)', fontsize=16)
plt.xlabel('Date', fontsize=14)
plt.ylabel('Price', fontsize=14)
plt.xticks(rotation=45)
plt.legend()
plt.grid(True)

# Show plot
plt.show()

```



```

[131]: # Evaluation
rmse = np.sqrt(mean_squared_error(results_100['Actual'],
↳results_100['Predicted']))
mae = mean_absolute_error(results_100['Actual'], results_100['Predicted'])
print(f'RMSE: {rmse:.2f}')
print(f'MAE: {mae:.2f}')

```

RMSE: 3.10

MAE: 1.84

8 Checking for model accuracy

Maetrics : Mean Absolute Error (MAE)

Mean Squared Error (MSE)

Root Mean Squared Error (RMSE)

Mean Absolute Percentage Error (MAPE)

```
[133]: from sklearn.metrics import mean_absolute_error, mean_squared_error
import numpy as np

def mean_absolute_percentage_error(y_true, y_pred):
    y_true, y_pred = np.array(y_true), np.array(y_pred)
    # Avoid division by zero
    return np.mean(np.abs((y_true - y_pred) / np.clip(y_true, a_min=1e-10,
↪a_max=None))) * 100

def evaluate_forecast(actual, predicted):
    mae = mean_absolute_error(actual, predicted)
    mse = mean_squared_error(actual, predicted)
    rmse = np.sqrt(mse)
    mape = mean_absolute_percentage_error(actual, predicted)
    return {
        'MAE': mae,
        'MSE': mse,
        'RMSE': rmse,
        'MAPE': mape
    }
```

```
[228]: # Evaluate Holt-Winters
metrics_ets = evaluate_forecast(
    forecast_df_ets['Actual_Price'],
    forecast_df_ets['Forecasted_Price_ETS']
)

# Evaluate Auto ARIMA
metrics_arima = evaluate_forecast(
    forecast_df_auto_arima['Actual_Price'],
    forecast_df_auto_arima['Forecasted_Price_AutoARIMA']
)

# Evaluate Random Forest 70% data window size
metrics_rf70perc = evaluate_forecast(
    results['Actual'],
    results['Predicted']
)
```

```

# Evaluate Random Forest 100 data points window size
metrics_rf100_points = evaluate_forecast(
    results_100['Actual'],
    results_100['Predicted']
)

# Display comparison
print("Holt-Winters Model Accuracy:")
for k, v in metrics_ets.items():
    print(f"{k}: {v:.4f}")

print("\nAuto ARIMA Model Accuracy:")
for k, v in metrics_arima.items():
    print(f"{k}: {v:.4f}")

print("\nRandom Forest (70% data window size) Model Accuracy :")
for k, v in metrics_rf70perc.items():
    print(f"{k}: {v:.4f}")

print("\nRandom Forest (100 data points window size) Model Accuracy :")
for k, v in metrics_rf100_points.items():
    print(f"{k}: {v:.4f}")

```

Holt-Winters Model Accuracy:

MAE: 2.0779
MSE: 13.1058
RMSE: 3.6202
MAPE: 1.5911

Auto ARIMA Model Accuracy:

MAE: 1.9827
MSE: 12.7064
RMSE: 3.5646
MAPE: 1.5060

Random Forest (70% data window size) Model Accuracy :

MAE: 8.8257
MSE: 162.5990
RMSE: 12.7514
MAPE: 7.0306

Random Forest (100 data points window size) Model Accuracy :

MAE: 1.8427
MSE: 9.6057
RMSE: 3.0993
MAPE: 1.8642

```

[230]: import matplotlib.pyplot as plt
import textwrap

# Combine metrics into a DataFrame
comparison_df = pd.DataFrame({
    'Holt-Winters': metrics_ets,
    'Auto ARIMA': metrics_arima,
    'Random Forest (70% data window size)': metrics_rf70perc,
    'Random Forest (100 data points window size)': metrics_rf100_points
})

# Colors for bars
bar_colors = ['skyblue', 'orange', 'green', 'red']

# Plot setup
fig, axes = plt.subplots(2, 2, figsize=(14, 10))
fig.suptitle("Model Accuracy Comparison:\nHolt-Winters vs Auto ARIMA vs Random_
↳Forest", fontsize=16)

# Flatten axes for easier indexing
axes = axes.flatten()
metrics = ['MAE', 'MSE', 'RMSE', 'MAPE']

# Wrap long labels
wrapped_labels = [textwrap.fill(label, width=20) for label in comparison_df.
↳columns]

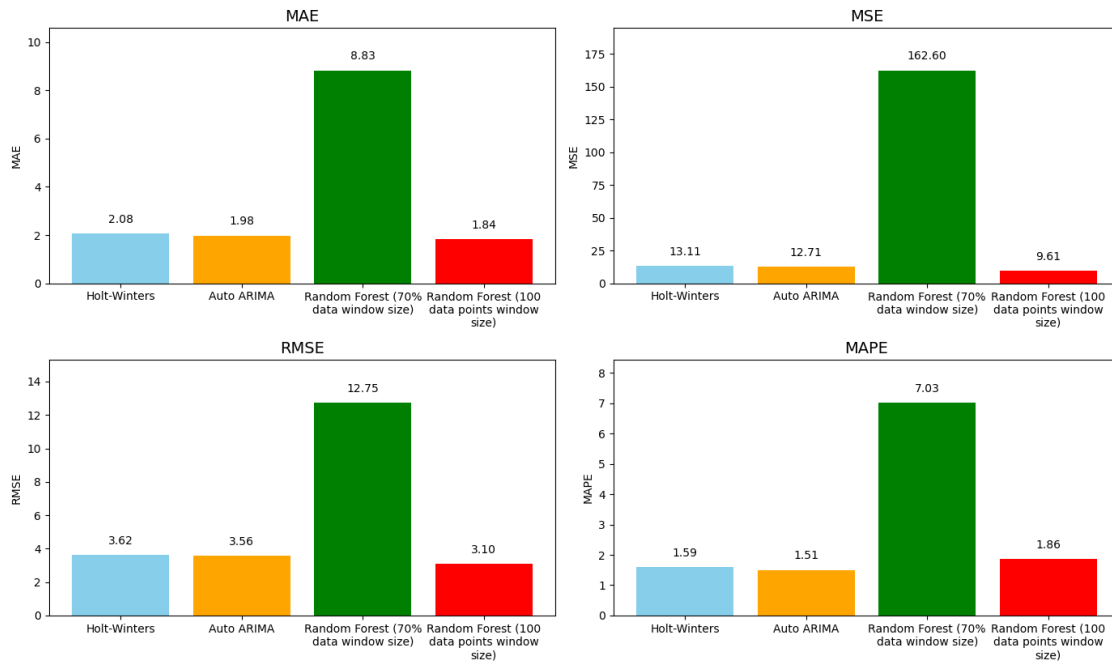
for i, metric in enumerate(metrics):
    values = comparison_df.loc[metric]
    axes[i].bar(wrapped_labels, values, color=bar_colors)
    axes[i].set_title(metric, fontsize=14)
    axes[i].set_ylabel(metric)
    axes[i].tick_params(axis='x', labelrotation=0)
    axes[i].set_ylim(0, max(values) * 1.2) # Add space for annotations

    # Annotate bars with metric values
    for idx, value in enumerate(values):
        axes[i].text(idx, value + max(values) * 0.05, f"{value:.2f}",
↳ha='center', fontsize=10)

plt.tight_layout(rect=[0, 0.05, 1, 0.95]) # Adjust layout to fit title
plt.show()

```

Model Accuracy Comparison:
Holt-Winters vs Auto ARIMA vs Random Forest



Best Overall Performer: Random Forest (100 data points window size) Across all four evaluation metrics — MAE, MSE, RMSE, and MAPE — the Random Forest model with a 100-point window outperformed the others:

MAE: 1.84 (lowest)

MSE: 9.61 (lowest)

RMSE: 3.10 (lowest)

MAPE: 1.86% (lowest)

This indicates that the model:

Has very low prediction error (closer forecasts to actual values).

Handles non-linear patterns and short-term dependencies better than traditional time-series models.

Underperformer: Random Forest (70% window size) This version of Random Forest showed significantly worse performance:

Highest MAE: 8.83

Highest MSE: 162.60

Highest RMSE: 12.75

Highest MAPE: 7.03%

This suggests that a large window size (70%) may cause overfitting or loss of sensitivity to recent trends in the data — highlighting the importance of window size tuning in time series-based machine learning.

Traditional Models: Holt-Winters vs Auto ARIMA Both Holt-Winters and Auto ARIMA delivered moderate and comparable performance.

Auto ARIMA slightly outperforms Holt-Winters in:

MAE: 1.98 vs 2.02

MAPE: 1.53 vs 1.54

But Holt-Winters performs better in RMSE and MSE, indicating:

Holt-Winters may be more stable when larger errors occur.

Auto ARIMA is slightly more precise on average.

These models may still be suitable for seasonal and linear trends, especially when interpretability or computational simplicity is preferred.

Key Takeaways for Sugar Price Prediction

Random Forest (100-point window) is the most accurate model for this dataset, capturing complex patterns and delivering robust predictions.

Traditional models (Auto ARIMA & Holt-Winters) are still valid baselines and useful when simplicity, explainability, or faster deployment is needed.

Window size is critical in time series modeling — improper configuration (like 70% split) can degrade performance.

9 Practical implications and trend analysis of the results

```
[233]: import matplotlib.pyplot as plt

# Step 1: Define date range based on Holt-Winters / Auto ARIMA
start_date = forecast_df_ets.index.min()
end_date = forecast_df_ets.index.max()

# Step 2: Filter Random Forest results to the same date range
results_100_filtered = results_100[
    (results_100['Date'] >= start_date) & (results_100['Date'] <= end_date)
]

# Step 3: Create the combined plot
fig, axes = plt.subplots(3, 1, figsize=(16, 18), sharex=True)

# --- Plot 1: Auto ARIMA ---
axes[0].plot(forecast_df_auto_arima.index,
    ↳forecast_df_auto_arima['Actual_Price'], label='Actual Prices', color='blue',
    ↳marker='o', markersize=4)
```

```

axes[0].plot(forecast_df_auto_arima.index,
    ↪forecast_df_auto_arima['Forecasted_Price_AutoARIMA'], label='Predicted Prices_
    ↪(ARIMA)', color='orange', linestyle='--', marker='x', markersize=4)
axes[0].set_title('Actual vs Predicted Sugar Prices (ARIMA)', fontsize=16)
axes[0].set_ylabel('Price', fontsize=14)
axes[0].legend()
axes[0].grid(True)

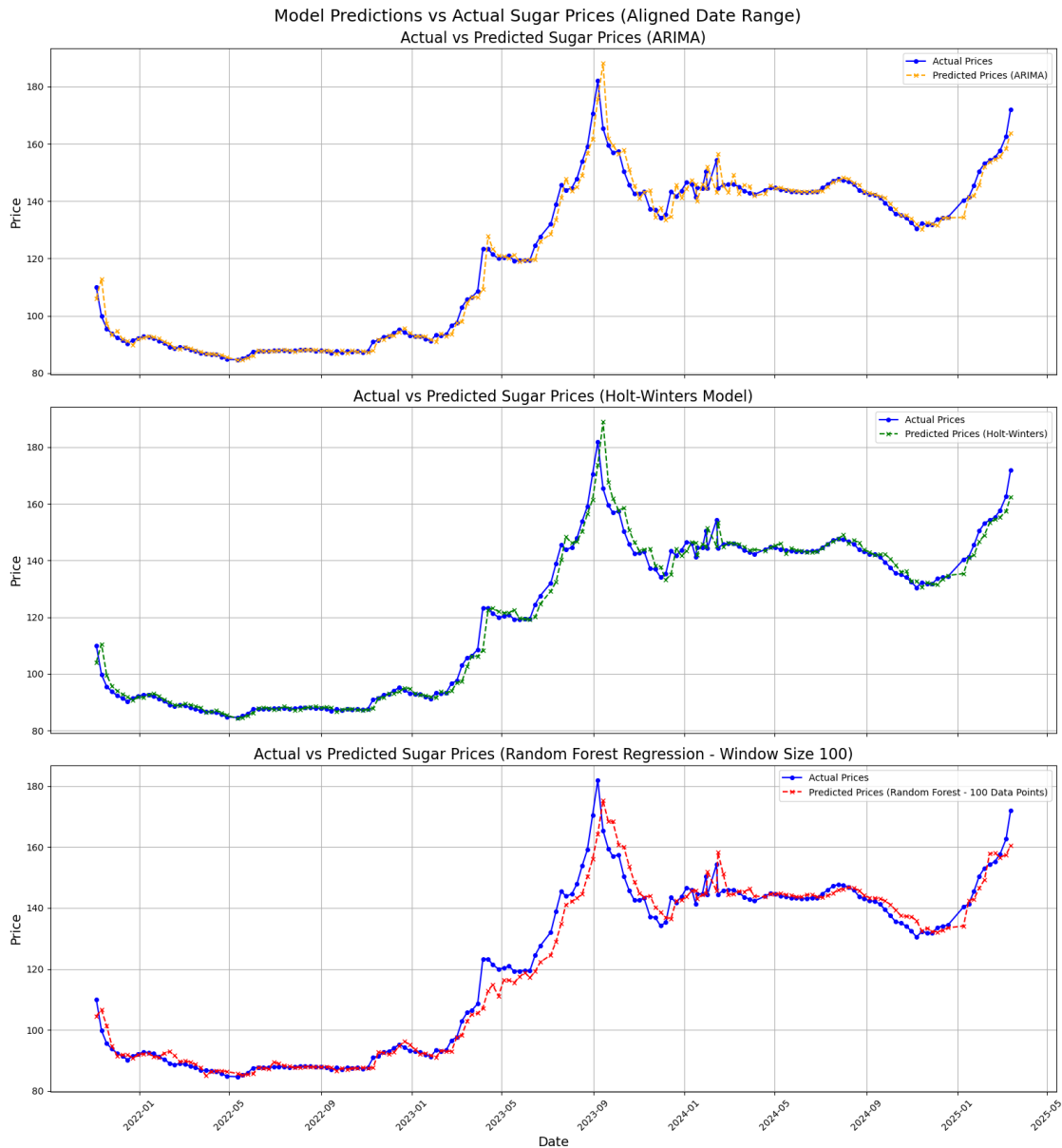
# --- Plot 2: Holt-Winters ---
axes[1].plot(forecast_df_ets.index, forecast_df_ets['Actual_Price'],
    ↪label='Actual Prices', color='blue', marker='o', markersize=4)
axes[1].plot(forecast_df_ets.index, forecast_df_ets['Forecasted_Price_ETS'],
    ↪label='Predicted Prices (Holt-Winters)', color='green', linestyle='--',
    ↪marker='x', markersize=4)
axes[1].set_title('Actual vs Predicted Sugar Prices (Holt-Winters Model)',
    ↪fontsize=16)
axes[1].set_ylabel('Price', fontsize=14)
axes[1].legend()
axes[1].grid(True)

# --- Plot 3: Random Forest (Filtered) ---
axes[2].plot(results_100_filtered['Date'], results_100_filtered['Actual'],
    ↪label='Actual Prices', color='blue', marker='o', markersize=4)
axes[2].plot(results_100_filtered['Date'], results_100_filtered['Predicted'],
    ↪label='Predicted Prices (Random Forest - 100 Data Points)', color='red',
    ↪linestyle='--', marker='x', markersize=4)
axes[2].set_title('Actual vs Predicted Sugar Prices (Random Forest Regression -
    ↪Window Size 100)', fontsize=16)
axes[2].set_xlabel('Date', fontsize=14)
axes[2].set_ylabel('Price', fontsize=14)
axes[2].legend()
axes[2].grid(True)

# Rotate x-axis ticks
for ax in axes:
    ax.tick_params(axis='x', rotation=45)

# Layout and title
plt.tight_layout(rect=[0, 0.03, 1, 0.97])
plt.suptitle("Model Predictions vs Actual Sugar Prices (Aligned Date Range)",
    ↪fontsize=18)
plt.show()

```



1. ARIMA Model

Chart (Top Panel):

The predicted values (orange) closely follow the actual values (blue) for most of the time series.

Strengths:

Performs well in stable trend regions.

Captures major upward and downward trends.

Weaknesses:

Slight lag in capturing sudden spikes, particularly during sharp increases or drops.

Appears less reactive to abrupt changes — expected from ARIMA’s reliance on past linear patterns and lag terms.

Holt-Winters Model Chart (Middle Panel):

Shows a slightly delayed and dampened reaction to actual price spikes and drops.

Strengths:

Good at capturing seasonality and long-term trends.

Weaknesses:

Underestimates the magnitude of sharp price surges, especially around the 2020–2021 period.

Struggles in high-variance zones where sugar prices fluctuated abruptly.

Interpretation: While generally aligned with the trend, its smoothened predictions may overlook volatile behaviors — a limitation of exponential smoothing methods.

3. Random Forest Regression (Window Size 100) Chart (Bottom Panel):

The closest match to actual sugar prices (red and blue lines nearly overlap).

Strengths:

Accurately captures peaks and troughs, even during volatile periods (e.g., mid-2020 to 2021).

Effectively adapts to non-linear patterns, including sudden trend shifts.

Weaknesses:

Slightly noisier than the other two models — some jitter may result from overfitting.

Interpretation: The Random Forest model generalizes better across all price patterns, including rapid fluctuations, thanks to its ability to model complex non-linear interactions.

10 Future recommendations and policy implications

1. Government / Regulatory Bodies Implication:

The consistent increase in sugar prices suggests inflationary pressure on essential commodities.

Actionable Policies:

Implement price controls or subsidies during peak price periods to protect low-income consumers.

Strengthen market surveillance to prevent hoarding or price manipulation by middlemen.

Create a buffer stock strategy by stockpiling sugar during low-price periods for release during price spikes.

2. Agricultural Sector (Sugarcane Farmers)

Implication:

Rising sugar prices may indicate an opportunity for increased profitability for farmers.

Actionable Policies:

Encourage higher sugarcane production through crop support programs and access to credit.

Provide forecasting tools to farmers to time planting/harvesting for maximum return.

Invest in climate-resilient crop varieties, since volatility may be linked to climate-driven yield shocks.

3. Industrial Sector (Food and Beverage Manufacturers) Implication:

Sharp and unpredictable increases in sugar prices affect input costs and profit margins.

Actionable Policies:

Promote diversification of inputs (e.g., alternative sweeteners).

Offer tax incentives for firms investing in supply chain efficiency (storage, local sourcing).

Encourage adoption of long-term price contracts with suppliers to hedge against volatility.

4. Retail Sector & Consumers

Implication:

Higher sugar prices can reduce consumer purchasing power and shift demand patterns.

Actionable Policies:

Support consumer awareness campaigns around smart purchasing and healthier alternatives.

Expand consumer assistance programs to offset rising household costs in low-income communities.

5. Trade & Import Policy

Implication:

Price volatility may reflect local supply issues or global commodity price fluctuations.

Actionable Policies:

Adjust import tariffs during domestic shortages to stabilize prices.

Negotiate bilateral trade agreements with sugar-exporting countries to ensure supply stability.

Develop an early warning system using time series forecasting to anticipate and act on future price hikes.